Refinement and Type Difference in SKJS

Abstract

We formalize the internal type representation used by SKJS and study two related type transformations: (1) predicate-driven *refinement* (control-flow narrowing), and (2) the type-difference operation (diff) used during type inference. We state and prove soundness properties for these transformations, while explicitly documenting the intended sources of incompleteness.

1 Definitions Index

Symbol	Defined In
Tags	Definition 1
$(\mathsf{Payload}_k)_{k\in\mathsf{Tags}}, \mathsf{Val}, \mathsf{rtag}$	Definition 2
SType	Definition 3
tag	Definition 4
A0	Lemma 5
A1	Lemma 6
$\mathsf{STypes}, \mathsf{tags}(\cdot)$	Definition 9
$\mathsf{STC}(\cdot,\cdot)$	Definition 47
TInst, TInsts, TParam	Definitions 13–15
Clause, Clauses	Definition 16
TType	Definition 17
refine (predicate-driven refinement)	Definition 21
Predicate-soundness	Definition 23
diff (soundness P1/P2)	Definition 42

Type Hierarchy and Notation Summary

Type representation hierarchy (each level builds on the previous):

$$\underbrace{\mathsf{SType}}_{\text{single tag}} \xrightarrow{\text{set}} \underbrace{\mathsf{STypes}}_{\text{tag-distinct}} \xrightarrow{+\text{ params}} \underbrace{\mathsf{Clause}}_{\langle V|S\rangle} \xrightarrow{\text{map}} \underbrace{\mathsf{Clauses}}_{V\mapsto\langle V|S\rangle} \xrightarrow{+\text{ insts}} \underbrace{\mathsf{TType}}_{(I,C)}$$

Semantic interpretation $[\![\cdot]\!] : X \to \mathcal{P}(\mathsf{Val})$:

$$[\![\mathsf{STypes}]\!] = \bigcup_{s\in S}[\![s]\!] \qquad [\![\langle V \mid S\rangle]\!] = \left(\bigcap_{v\in V}[\![v]\!]\right) \cap [\![S]\!] \qquad [\![(I,C)]\!] = [\![I]\!] \cup [\![C]\!]$$

Soundness contracts:

- *Predicate soundness* (refinement): $[\![t]\!] \cap P \subseteq [\![R(t)]\!]$
- *Diff soundness* (P1+P2): $[\![t_1]\!] \setminus [\![t_2]\!] \subseteq [\![\mathsf{diff}(t_1,t_2)]\!] \subseteq [\![t_1]\!]$

2 Runtime Model and Semantic Interface

Definition 1 (Runtime Tags)**.** *Let* Tags *be the set of runtime tags (the coarse runtime kind of a value). In the current implementation, tags are represented as integers, so one may take* $\mathsf{Tags} = \{0, 1, \ldots, 10\}$.

Definition 2 (Runtime Values and Tags)**.** *Assume runtime values carry their tag as part of the representation. We fix a family of payload sets* $(\mathsf{Payload}_k)_{k\in\mathsf{Tags}}$ *and define*

$$\mathsf{Val} \triangleq \{\, (k,p) \mid k \in \mathsf{Tags} \ \wedge \ p \in \mathsf{Payload}_k \,\}.$$

Define the runtime tag function $\mathsf{rtag} : \mathsf{Val} \to \mathsf{Tags}$ *as the first projection:*

$$\mathsf{rtag}((k,p)) = k.$$

Concrete payloads (for this document). *We make primitive payloads concrete and leave structured payloads abstract:*

$$\mathsf{Payload}_0 = \{\star\} \qquad \mathsf{Payload}_1 = \{\star\} \qquad \mathsf{Payload}_2 = \{\star\}$$
$$\mathsf{Payload}_3 = \{\mathsf{true}, \mathsf{false}\} \qquad \mathsf{Payload}_4 = \mathbb{R} \qquad \mathsf{Payload}_5 = \mathsf{String}$$
$$\mathsf{Payload}_6 = \mathbb{Z} \qquad \mathsf{Payload}_{10} = \mathsf{String} \times \mathsf{Payload}'_{10}.$$

Here $\mathsf{Payload}'_{10}$ *is an arbitrary set (the constructor payload). For the remaining tags* $k \in \{7, 8, 9\}$ *we leave* $\mathsf{Payload}_k$ *as arbitrary sets.*

Definition 3 (Simple Types)**.** *Let* SType *be the set of* simple types*, i.e. the atomic type forms that have a single runtime tag. Formally,* SType *is the smallest set generated by the following constructors:*

- *primitive units:* `undefined`, `void`, `null`

- *primitive values with optional finite literal-set payload:*
  - $\texttt{boolean}(B)$ *where* $B \subseteq_{\text{fin}} \{\mathsf{true}, \mathsf{false}\}$ *or* $B = \top$,
  - $\texttt{number}(N)$ *where* $N \subseteq_{\text{fin}} \mathbb{R}$ *or* $N = \top$,
  - $\texttt{string}(S)$ *where* $S \subseteq_{\text{fin}} \mathsf{String}$ *or* $S = \top$,
  - $\texttt{bigint}(Z)$ *where* $Z \subseteq_{\text{fin}} \mathbb{Z}$ *or* $Z = \top$.
- *structured atoms:* $\texttt{tuple}(\ldots)$, $\texttt{Promise}(\ldots)$, $\texttt{object}(\ldots)$, $\texttt{algebraic}(\ldots)$. *These include tagged unions/ADTs with a runtime discriminant.*

We write these constructors suggestively; most internal parameters are left abstract. However, for control-flow refinement we will require a concrete semantic interface for object shapes (fields and index signatures); see Definition 7.

Notation (source-level literal types). *We write* $\texttt{boolean}$ *for* $\texttt{boolean}(\top)$*, and similarly for* $\texttt{number}$, $\texttt{string}$, $\texttt{bigint}$. *We write* $\texttt{true}$ *for* $\texttt{boolean}(\{\mathsf{true}\})$ *and* $\texttt{false}$ *for* $\texttt{boolean}(\{\mathsf{false}\})$. *We write a numeric literal* n *for* $\texttt{number}(\{n\})$*, and a string literal "*x*" for* $\texttt{string}(\{x\})$. *More generally, a union like* `1 | 2` *is represented as* $\texttt{number}(\{1, 2\})$*, i.e. a single* SType *alternative at the numeric tag. We use these abbreviations uniformly, including under* $\mathsf{tag}(\cdot)$ *and* $[\![\cdot]\!]$. *For example,* $[\![\texttt{number}]\!]$ *means* $[\![\texttt{number}(\top)]\!]$*, and* $\mathsf{tag}(\texttt{1})$ *means* $\mathsf{tag}(\texttt{number}(\{1\}))$.

Definition 4 (Simple Types and Tags)**.** *The function* $\mathsf{tag} : \mathsf{SType} \to \mathsf{Tags}$ *assigns tags:*

$$\begin{aligned}
\mathsf{tag}(\texttt{undefined}) &= 0 & \mathsf{tag}(\texttt{void}) &= 1 \\
\mathsf{tag}(\texttt{null}) &= 2 & \mathsf{tag}(\texttt{boolean}(\cdot)) &= 3 \\
\mathsf{tag}(\texttt{number}(\cdot)) &= 4 & \mathsf{tag}(\texttt{string}(\cdot)) &= 5 \\
\mathsf{tag}(\texttt{bigint}(\cdot)) &= 6 & \mathsf{tag}(\texttt{tuple}(\ldots)) &= 7 \\
\mathsf{tag}(\texttt{Promise}(\ldots)) &= 8 & \mathsf{tag}(\texttt{object}(\ldots)) &= 9 \\
\mathsf{tag}(\texttt{algebraic}(\ldots)) &= 10
\end{aligned}$$

The semantic interpretation $[\![\cdot]\!] : \mathsf{SType} \to \mathcal{P}(\mathsf{Val})$ *maps each simple type to its set of inhabitants. We define* $[\![\cdot]\!]$ *concretely for primitive units and primitive-value payloads (including singleton-literal notation), and define the remaining structured cases at tag level (with the* $\texttt{object}$ *and* $\texttt{algebraic}$ *cases further constrained in Definitions 7 and 8):*

$$\begin{aligned}
[\![\texttt{undefined}]\!] &\triangleq \{(0, \star)\} & [\![\texttt{void}]\!] &\triangleq \{(1, \star)\} & [\![\texttt{null}]\!] &\triangleq \{(2, \star)\} \\
[\![\texttt{boolean}(\top)]\!] &\triangleq \{(3, b) \mid b \in \{\mathsf{true}, \mathsf{false}\}\} & [\![\texttt{boolean}(B)]\!] &\triangleq \{(3, b) \mid b \in B\} \\
[\![\texttt{number}(\top)]\!] &\triangleq \{(4, n) \mid n \in \mathbb{R}\} & [\![\texttt{number}(N)]\!] &\triangleq \{(4, n) \mid n \in N\} \\
[\![\texttt{string}(\top)]\!] &\triangleq \{(5, x) \mid x \in \mathsf{String}\} & [\![\texttt{string}(S)]\!] &\triangleq \{(5, x) \mid x \in S\} \\
[\![\texttt{bigint}(\top)]\!] &\triangleq \{(6, z) \mid z \in \mathbb{Z}\} & [\![\texttt{bigint}(Z)]\!] &\triangleq \{(6, z) \mid z \in Z\} \\
[\![\texttt{tuple}(\ldots)]\!] &\triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = 7\} \\
[\![\texttt{Promise}(\ldots)]\!] &\triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = 8\}.
\end{aligned}$$

In the finite-payload cases, B, N, S, Z *denote finite subsets of the respective domains.*

Lemma 5 (**A0** (Runtime Tag Consistency))**.** *The semantic interpretation of simple types agrees with runtime tags:*

$$\forall s \in \mathsf{SType}.\ \forall v \in \mathsf{Val}.\ v \in [\![s]\!] \implies \mathsf{rtag}(v) = \mathsf{tag}(s).$$

Proof. By case analysis on s. For primitive units and primitive-value payloads, the definition of $[\![s]\!]$ contains only values of the form $(\mathsf{tag}(s), p)$, so $\mathsf{rtag}(v) = \mathsf{tag}(s)$. For the tag-level structured cases (`tuple`, `Promise`), $[\![s]\!]$ is defined as $\{v \mid \mathsf{rtag}(v) = k\}$ where $k = \mathsf{tag}(s)$. For `object` and `algebraic`, Definitions 7 and 8 explicitly require membership in ObjVal (tag 9) and AlgVal (tag 10) respectively, and these tags agree with tag. □

Lemma 6 (**A1** (Tag Disjointness))**.**

$$\forall s_1, s_2 \in \mathsf{SType}.\ \mathsf{tag}(s_1) \neq \mathsf{tag}(s_2) \implies [\![s_1]\!] \cap [\![s_2]\!] = \emptyset$$

Proof. Let $k_1 = \mathsf{tag}(s_1)$ and $k_2 = \mathsf{tag}(s_2)$ with $k_1 \neq k_2$. By Lemma 5, every $v \in [\![s_1]\!]$ satisfies $\mathsf{rtag}(v) = k_1$ and every $v \in [\![s_2]\!]$ satisfies $\mathsf{rtag}(v) = k_2$. No value can have two different tags, so $[\![s_1]\!] \cap [\![s_2]\!] = \emptyset$. □

Definition 7 (Object Values and Object-Type Semantics)**.** *Let* $\mathsf{ObjVal} \subseteq \mathsf{Val}$ *be the set of object-tagged runtime values:*

$$\mathsf{ObjVal} \triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = 9\}.$$

Assume a (partial) finite-property interface on objects:

$$\mathsf{props} : \mathsf{ObjVal} \to \mathcal{P}_{\mathrm{fin}}(\mathsf{String})$$
$$\mathsf{get} : \mathsf{ObjVal} \times \mathsf{String} \rightharpoonup \mathsf{Val}$$

such that $\mathsf{get}(o, p)$ *is defined exactly when* $p \in \mathsf{props}(o)$*. Write* $\mathsf{has}(o, p)$ *for* $p \in \mathsf{props}(o)$*.*

Object field specifications. *Assume predicates* $\mathsf{callable}, \mathsf{constructible} : \mathsf{ObjVal} \to \{\mathsf{true}, \mathsf{false}\}$ *testing whether an object value is callable or constructible.*

Three-valued signature modes. *To support precise refinement after normalization, we use a three-valued mode for call and constructor signatures:*

$$\mathsf{SigMode} \triangleq \{\mathsf{None},\ \mathsf{Maybe}(\sigma),\ \mathsf{Definite}(\sigma)\}$$

where σ *ranges over abstract signatures (we treat signatures opaquely in this document). Intuitively:*

- $\mathsf{Definite}(\sigma)$*: the object is definitely callable/constructible with signature* σ*.*
- $\mathsf{Maybe}(\sigma)$*: the object may or may not be callable/constructible; if it is, it has signature* σ*.*
- None*: the object is definitely not callable/constructible.*

An object field specification *is a tuple* $F = (F_{\mathrm{req}}, F_{\mathrm{opt}}, F_{\mathrm{idx}}, F_{\mathrm{new}}, F_{\mathrm{call}})$ *where:*

- $F_{\mathrm{req}} : \mathsf{String} \rightharpoonup \mathsf{TType}$ *is a finite partial map of required fields,*
- $F_{\mathrm{opt}} : \mathsf{String} \rightharpoonup \mathsf{TType}$ *is a finite partial map of optional fields,*
- F_{idx} *is either* None *or* $\mathsf{Some}(K, V)$ *for* $K, V \in \mathsf{TType}$ *(an index signature),*
- $F_{\mathrm{new}} \in \mathsf{SigMode}$ *for constructor signatures,*
- $F_{\mathrm{call}} \in \mathsf{SigMode}$ *for call signatures.*

Semantics of `object`(F)**.** *We assume the object constructor in* SType *can carry such an* F*, and define:*

$$o \in [\![\texttt{object}(F)]\!]$$

if and only if $o \in \mathsf{ObjVal}$ *and the following constraints hold:*

- ***Required fields:*** *for every* $p \in \mathrm{dom}(F_{\mathrm{req}})$*,* $\mathsf{has}(o,p)$ *and* $\mathsf{get}(o,p) \in [\![F_{\mathrm{req}}(p)]\!]$*.*
- ***Optional fields:*** *for every* $p \in \mathrm{dom}(F_{\mathrm{opt}})$*, if* $\mathsf{has}(o,p)$ *then* $\mathsf{get}(o,p) \in [\![F_{\mathrm{opt}}(p)]\!]$ *(and if not, no constraint).*
- ***Index signature (open objects):*** *if* $F_{\mathrm{idx}} = \mathsf{Some}(K,V)$ *then for every* $p \in \mathsf{props}(o)$*: if the runtime string value "*p*" is in* $[\![K]\!]$*, then* $\mathsf{get}(o,p) \in [\![V]\!]$*.*
- ***Constructor signature:***
  - *if* $F_{\mathrm{new}} = \mathsf{Definite}(\nu)$ *then* $\mathsf{constructible}(o) = \mathsf{true}$*;*
  - *if* $F_{\mathrm{new}} = \mathsf{Maybe}(\nu)$ *then no constraint (object may or may not be constructible);*
  - *if* $F_{\mathrm{new}} = \mathsf{None}$ *then* $\mathsf{constructible}(o) = \mathsf{false}$*.*
- ***Call signature:***
  - *if* $F_{\mathrm{call}} = \mathsf{Definite}(\kappa)$ *then* $\mathsf{callable}(o) = \mathsf{true}$*;*
  - *if* $F_{\mathrm{call}} = \mathsf{Maybe}(\kappa)$ *then no constraint (object may or may not be callable);*
  - *if* $F_{\mathrm{call}} = \mathsf{None}$ *then* $\mathsf{callable}(o) = \mathsf{false}$*.*

Note. *This semantics is intentionally* open*: objects may have properties not mentioned in* F_{req} *or* F_{opt}*. The three-valued signature modes ensure that normalization (merging callable with non-callable objects) is sound, while still permitting precise refinement via* `typeof`*.*

Definition 8 (Algebraic Values and Tag-Based Algebraic-Type Semantics)**.** *Let* $\mathsf{AlgVal} \subseteq \mathsf{Val}$ *be the set of algebraic-tagged runtime values:*

$$\mathsf{AlgVal} \triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = 10\}.$$

Recall that $\mathsf{Payload}_{10} = \mathsf{String} \times \mathsf{Payload}'_{10}$ *(Definition 2). Define the runtime discriminant function* $\mathsf{atag} : \mathsf{AlgVal} \to \mathsf{String}$ *by projection:*

$$\mathsf{atag}((10,(\ell,p))) \triangleq \ell.$$

Semantics of `algebraic`(T)**.** *We assume the algebraic constructor in* SType *can carry a finite set of tags* $T \subseteq_{\mathrm{fin}} \mathsf{String}$*, and define:*

$$v \in [\![\texttt{algebraic}(T)]\!] \quad \textit{iff} \quad v \in \mathsf{AlgVal} \ \wedge \ \mathsf{atag}(v) \in T.$$

Note. *This models only the discriminant-level behavior needed for refinement; field/constructor payload constraints are left abstract.*

3 Internal Type Representation

Definition 9 (STypes)**.**

> STypes *is a finite set* $S \subseteq_{\text{fin}} \mathsf{SType}$ *with pairwise distinct tags.*
> ***Semantics:*** $[\![S]\!] = \bigcup_{s \in S} [\![s]\!]$.

1. ***Tag-distinctness:*** $s_1 \neq s_2 \implies \mathsf{tag}(s_1) \neq \mathsf{tag}(s_2)$.
2. ***Lookup:*** $\mathsf{tags}(S) = \{\mathsf{tag}(s) \mid s \in S\}$. *Write* $S[k]$ *for the unique* $s \in S$ *with* $\mathsf{tag}(s) = k$.
3. ***Intuition:*** *An* STypes *value represents the* tag-level *part of a union type—one representative* SType *per runtime tag.*
4. ***Example:*** `number | string` *is* $\{$`number`, `string`$\}$. *But* `number` *and literal* `1` *share tag* 4*, so they cannot both appear in an* STypes*.*

3.1 Normal-Form Operations on STypes

Operations (API). We treat STypes as a normal-form ADT and highlight the operations used by the implementation:

- **Constructors:** $\emptyset \in \mathsf{STypes}$ and $\{s\} \in \mathsf{STypes}$ for $s \in \mathsf{SType}$.
- **Filter-map (refinement primitive):** for tag-preserving g and filter f,

$$\mathsf{filterMap}^{f,g}(S) \triangleq \{\, g(s) \mid s \in S,\ f(s) = \mathsf{true} \,\}.$$

- **Union (implementation union):** $S_1 \sqcup_{\mathsf{STypes}} S_2$ is the normal-form union operation used by SKJS, which merges same-tag alternatives into a single representative per tag (while keeping other tags unchanged).
- **Intersection (over-approximate meet):** the implementation provides a partial operation $S_1 \sqcap_{\mathsf{STypes}} S_2$ (it may return no result when the intersection is bottom). We model it abstractly below via its soundness property.
- **Difference (tag-elimination):** $\mathsf{diff}_{\mathsf{STypes}}(S_1, S_2)$ is defined in Section 45.

Soundness summary. All operations above are intended to be sound with respect to $[\![\cdot]\!]$:

- **Empty/singleton (exact):** $[\![\emptyset]\!] = \emptyset$ and $[\![\{s\}]\!] = [\![s]\!]$.
- **Filter-map (predicate-sound):** if (f, g) is sound for predicate P (Definition 21) and g preserves tags, then

$$[\![S]\!] \cap P \subseteq [\![\mathsf{filterMap}^{f,g}(S)]\!].$$

- **Union (over-approximate union):**

$$[\![S_1]\!] \cup [\![S_2]\!] \subseteq [\![S_1 \sqcup_{\mathsf{STypes}} S_2]\!].$$

- **Intersection (over-approximate intersection):** when $S_1 \sqcap_{\mathsf{STypes}} S_2$ is defined,

$$[\![S_1]\!] \cap [\![S_2]\!] \subseteq [\![S_1 \sqcap_{\mathsf{STypes}} S_2]\!].$$

- **Difference:** (P1) always holds (Lemma 46); (P2) holds under STC (Lemma 48).

Lemma 10 (Union Soundness for STypes)**.** *For all $S_1, S_2 \in \mathsf{STypes}$,*

$$[\![S_1]\!] \cup [\![S_2]\!] \subseteq [\![S_1 \sqcup_{\mathsf{STypes}} S_2]\!].$$

Proof. We will define $S_1 \sqcup_{\mathsf{STypes}} S_2$ to match the implementation's tag-directed union operation (Definition 11), and soundness follows from the corresponding same-tag merge soundness of each case. □

Definition 11 (Implementation Union/Intersection on STypes)**.** *We define $\sqcup_{\mathsf{STypes}}$ and $\sqcap_{\mathsf{STypes}}$ to model SKJS's operations on normal-form tag-distinct unions.*

Same-tag merge on simple types. *Define partial operators $\sqcup_{\mathsf{SType}}$ and $\sqcap_{\mathsf{SType}}$ on* SType*, used only when $\mathsf{tag}(s_1) = \mathsf{tag}(s_2)$. They follow the implementation's tag-dispatch (see `SType::union` and `SType::inter`):*

- ***Primitive units (tags 0/1/2):*** *$s_1 \sqcup_{\mathsf{SType}} s_2 \triangleq s_1$ and $s_1 \sqcap_{\mathsf{SType}} s_2 \triangleq s_1$. The implementation keeps the left operand; semantics is unchanged.*
- ***Primitive values (tags 3/4/5/6):*** *$\sqcup_{\mathsf{SType}}$ computes the set union of literal-sets when both are finite, and otherwise yields the full primitive kind (e.g. if either operand is $\top$, the result is $\top$). For booleans, unioning $\{\mathsf{true}\}$ and $\{\mathsf{false}\}$ yields `boolean`$(\top)$. Intersection $\sqcap_{\mathsf{SType}}$ keeps only common literals, or is undefined if none.*
- ***Tuples (tag 7):*** *$\sqcup_{\mathsf{SType}}$ unions element-wise up to the shorter length; missing positions in the shorter tuple are treated as `undefined` and unioned with the corresponding element of the longer tuple. $\sqcap_{\mathsf{SType}}$ intersects element-wise up to the shorter length and appends the remainder of the longer tuple unchanged.*
- ***Promises (tag 8):*** *`Promise`$(t_1) \sqcup_{\mathsf{SType}}$ `Promise`$(t_2) \triangleq$ `Promise`$(t_1 \sqcup_{\mathsf{TType}} t_2)$, and similarly $\sqcap_{\mathsf{SType}}$ uses $\sqcap_{\mathsf{TType}}$ (left abstract here).*
- ***Objects (tag 9):*** *For `object`$(F_1) \sqcup_{\mathsf{SType}}$ `object`(F_2):*
  - Fields: *union shared field types; make non-shared fields optional.*
  - Index signatures: *union key/value types into a single index signature.*
  - Call/new signatures: *keep only if both sides have one; otherwise drop.*

  Intersection is defined analogously and may be undefined (bottom) in some cases.
- ***Algebraic (tag 10):*** *`algebraic`$(T_1) \sqcup_{\mathsf{SType}}$ `algebraic`(T_2) unions the constructor-tag set; when a constructor exists on both sides, it merges its field specifications using object-field union. $\sqcap_{\mathsf{SType}}$ intersects constructor-tag sets and merges shared constructors; it is undefined if the tag-set intersection is empty.*

Union on STypes. *For $S_1, S_2 \in \mathsf{STypes}$, define:*

$$\begin{aligned} S_1 \sqcup_{\mathsf{STypes}} S_2 \triangleq\ & \{\, S_1[k] \mid k \in \mathsf{tags}(S_1) \setminus \mathsf{tags}(S_2) \,\} \\ & \cup \{\, S_2[k] \mid k \in \mathsf{tags}(S_2) \setminus \mathsf{tags}(S_1) \,\} \\ & \cup \{\, S_1[k] \sqcup_{\mathsf{SType}} S_2[k] \mid k \in \mathsf{tags}(S_1) \cap \mathsf{tags}(S_2) \,\}. \end{aligned}$$

Intersection on STypes. *Define the (partial) intersection:*

$$S_1 \sqcap_{\mathsf{STypes}} S_2 \triangleq \{\, S_1[k] \sqcap_{\mathsf{SType}} S_2[k] \mid k \in \mathsf{tags}(S_1) \cap \mathsf{tags}(S_2) \text{ and } S_1[k] \sqcap_{\mathsf{SType}} S_2[k] \text{ defined} \,\}.$$

If this set is empty, we treat the result as undefined (bottom), matching the implementation behavior.

Lemma 12 (Soundness of $\sqcup_{\mathsf{STypes}}$ and $\sqcap_{\mathsf{STypes}}$)**.** *For all $S_1, S_2 \in \mathsf{STypes}$:*

$$[\![S_1]\!] \cup [\![S_2]\!] \subseteq [\![S_1 \sqcup_{\mathsf{STypes}} S_2]\!].$$

Moreover, when $S_1 \sqcap_{\mathsf{STypes}} S_2$ is defined,

$$[\![S_1]\!] \cap [\![S_2]\!] \subseteq [\![S_1 \sqcap_{\mathsf{STypes}} S_2]\!].$$

Proof. For $\sqcup_{\mathsf{STypes}}$: let $v \in [\![S_1]\!] \cup [\![S_2]\!]$. WLOG $v \in [\![S_1]\!]$, so $v \in [\![S_1[k]]\!]$ for some $k \in \mathsf{tags}(S_1)$. If $k \notin \mathsf{tags}(S_2)$, then $S_1[k] \in S_1 \sqcup_{\mathsf{STypes}} S_2$, hence $v \in [\![S_1 \sqcup_{\mathsf{STypes}} S_2]\!]$. If $k \in \mathsf{tags}(S_2)$, then $S_1[k] \sqcup_{\mathsf{SType}} S_2[k] \in S_1 \sqcup_{\mathsf{STypes}} S_2$, and inclusion reduces to the same-tag merge soundness of $\sqcup_{\mathsf{SType}}$ for that tag case.

For $\sqcap_{\mathsf{STypes}}$: let $v \in [\![S_1]\!] \cap [\![S_2]\!]$. Then $v \in [\![S_1[k]]\!]$ and $v \in [\![S_2[k]]\!]$ for the same tag k by A1, and inclusion reduces to the same-tag merge soundness of $\sqcap_{\mathsf{SType}}$ when defined. □

3.2 Completeness and Precision Loss in Normal-Form Operations

We say an operation is *complete* when its result is semantically exact (i.e. no extra values are introduced), and *imprecise* when it introduces a strict over-approximation. This section identifies the specific implementation-driven steps where over-approximation is introduced, to inform potential future precision warnings.

Union is exact for primitive tags. For primitive units and primitive values, $\sqcup_{\mathsf{STypes}}$ is complete: it computes exactly the semantic union of the two alternatives at those tags. (E.g. numeric/string/bigint literal-sets are explicitly stored and unioned; boolean literals union to `boolean`.)

Union is generally imprecise for structured tags. For tuples, objects, promises, and algebraics, the same-tag merge rules may introduce extra inhabitants. Key sources of imprecision:

- **Object field merging via optionalization:** Merging $\texttt{object}(\{a : \tau\})$ and $\texttt{object}(\{b : \upsilon\})$ yields an object type where both a and b are optional. This admits objects containing *both* fields, which need not be in the original disjoint union. This is the primary, intentional precision loss caused by STypes storing at most one object alternative per tag.
- **Index signature union:** When both sides have index signatures, SKJS unions key/value types into a single index signature, losing per-key distinctions (see `TIndexField::union`).
- **Call/new signature merge:** When only one side has a call (or new) signature, SKJS drops it in the merged object. This loses the ability to distinguish callable from non-callable alternatives inside a single object tag.
- **Tuple union padding with `undefined`:** When tuple lengths differ, missing elements in the shorter tuple are unioned with `undefined`, potentially admitting tuples with explicit `undefined` entries not present in the shorter alternative.

Intersection imprecision for structured tags. As with union, $\sqcap_{\mathsf{STypes}}$ is exact for primitive units and primitive values (it intersects literal-sets precisely when they are tracked), but it can introduce over-approximation for structured tags:

- **Index signature intersection uses union (over-approximation):** When intersecting two index signatures, the implementation returns a *single* index signature whose key and value types are *unioned* (see `TIndexField::inter`). This is sound but can admit extra key/value pairs that satisfy neither original signature.
- **Tuple intersection on unequal lengths:** The implementation intersects element-wise up to the shorter length and appends the remaining elements of the longer tuple unchanged. This can admit tuples not in the exact set-theoretic intersection.

Clause-level intersection imprecision. At the clause level (not STypes), `Clauses::inter` accumulates clause-pair intersections in a map keyed by the variable-set. When multiple intersections land on the same key, they are combined using `Clause::union`, widening the result beyond the exact intersection in exchange for keeping at most one clause per key.

Difference is intentionally incomplete at same-tag. $\mathsf{diff}_{\mathsf{STypes}}$ removes *whole tags*. Therefore it cannot express same-tag subtraction such as `number` $\setminus$ `1`, which is why (P2) requires the same-tag coverage side condition STC (Section 47).

Filter-map precision loss. $\mathsf{filterMap}^{f,g}$ can lose precision whenever the filter drops alternatives or whenever the mapper performs a conservative rewrite. In SKJS, this shows up in refinements that (a) retain extra tags conservatively (e.g. `"in"` keeps tuples/promises/algebraics), or (b) perform only tag-level filtering without tracking richer correlations between fields.

Definition 13 (Type Instances)**.** *Let* TInst *be the set of type instances (type variables or generic instantiations).*

Examples. *A type variable instance is something like T (a unification variable introduced during type checking; this is* not *a source-level type parameter). A generic instantiation instance is a named type applied to type arguments, e.g.* `Promise`$\langle$`number`$\rangle$ *or* `Array`$\langle$`string`$\rangle$*.*

Each instance $i \in \mathsf{TInst}$ has a semantic interpretation $[\![i]\!] \subseteq \mathsf{Val}$.

Definition 14 (Type Instance Sets)**.** *Let* $\mathsf{TInsts} = \mathcal{P}_{\text{fin}}(\mathsf{TInst})$ *be the set of* finite *sets of instances. For $I \in \mathsf{TInsts}$:*

$$[\![I]\!] = \bigcup_{i \in I} [\![i]\!].$$

Definition 15 (Type Parameters)**.** *Let* TParam *be the set of* type parameters *(generic binders), i.e. names introduced by source-level generic declarations such as* `function f`$\langle T \rangle(\ldots)$ *or* `type Box`$\langle T \rangle = \ldots$*. Type parameters differ from type-variable instances: parameters are binders whose meaning is fixed by the caller, whereas instances are inference unknowns solved by the type checker.*

Definition 16 (Clause and Clauses)**.**

A clause $\langle V \mid S \rangle$ *pairs type parameters $V \subseteq \mathsf{TParam}$ with $S \in \mathsf{STypes} \cup \{\top\}$.*
Clauses *is a map from finite subsets of* TParam *to clauses, at most one per key.*

1. ***Notation**: We write $\langle V \mid S \rangle$ for a clause with parameters V and simple-types S. The symbol $\top$ denotes the "top" case (no type constraint); $\langle \emptyset \mid \top \rangle$ is the unique top clause.*

2. ***Semantics**:*

$$[\![\langle V \mid S \rangle]\!] = \left(\bigcap_{v \in V} [\![v]\!]\right) \cap [\![S]\!] \qquad \text{where } [\![\top]\!] = \mathsf{Val}$$
$$[\![C]\!] = \bigcup_{k \in \mathrm{dom}(C)} [\![C(k)]\!]$$

3. ***Well-formedness**: Keys agree with clause parameters: $C(k) = \langle V \mid S \rangle \implies V = k$. Top implies empty parameters: $S = \top \implies V = \emptyset$.*

4. ***Example**: Let $A, B \in \mathsf{TParam}$. The clauses map $C = \{A \mapsto \langle \{A\} \mid \texttt{number} \rangle,\ B \mapsto \langle \{B\} \mid \texttt{string} \rangle\}$ has:*

$$[\![C]\!] = ([\![A]\!] \cap [\![\texttt{number}]\!]) \cup ([\![B]\!] \cap [\![\texttt{string}]\!]).$$

A clause $\langle \{A, B\} \mid \texttt{number} \rangle$ requires both *parameters: $[\![A]\!] \cap [\![B]\!] \cap [\![\texttt{number}]\!]$.*

Definition 17 (TType)**.** $\mathsf{TType} = (\mathsf{insts}, \mathsf{clauses})$ *where* $\mathsf{insts} \in \mathsf{TInsts}$ *and* $\mathsf{clauses} \in \mathsf{Clauses}$.

$$[\![(\mathsf{insts}, \mathsf{clauses})]\!] = [\![\mathsf{insts}]\!] \cup [\![\mathsf{clauses}]\!]$$

Schematic. *It is helpful to keep the following picture in mind:*

$$\underbrace{(\mathsf{insts}, \mathsf{clauses})}_{\mathsf{TType}} \quad = \quad \underbrace{\mathsf{insts}}_{\text{“instance” part}} \quad \cup \quad \underbrace{\mathsf{clauses}}_{\text{“conditional” part}}$$

and later $\mathsf{diff}_{\mathsf{TType}}$ *simply applies* diff *componentwise to these two parts.*

Example. *Let $I = \{\texttt{Promise}\langle\texttt{number}\rangle\} \in \mathsf{TInsts}$ and let C be the clauses map from the previous example. Then $t = (I, C)$ denotes the type whose semantics is:*

$$[\![t]\!] = [\![\texttt{Promise}\langle\texttt{number}\rangle]\!] \cup [\![C]\!].$$

4 Subtyping (Skeleton)

We introduce a subtyping judgement on internal types, written $t_1 \leq t_2$ for $t_1, t_2 \in \mathsf{TType}$. In this document, $\leq$ is intended to be a faithful mathematical model of the subtyping *validator* implemented by SKJS (the accept/reject check used by the type checker), rather than an abstract, complete, semantic inclusion relation.

Soundness theorem (to be proved). The central meta-theoretic property we ultimately want from the validator-model judgement $\leq$ is semantic soundness:

$$t_1 \leq t_2 \quad \implies \quad [\![t_1]\!] \subseteq [\![t_2]\!].$$

In a full development, this is proved from the subtyping rules/algorithm that define $\leq$.

Incompleteness. The converse implication is intentionally *not* assumed: SKJS may reject some valid semantic inclusions, so $\leq$ can be incomplete.

Note (expansion and substitution effects). In the implementation, the validator operates on expanded/normalized views of types using the current substitution and a variance parameter, and it may update the substitution while checking (as part of inference). We do not model these mechanisms explicitly here; they are absorbed into the meaning of the judgement $t_1 \leq t_2$.

Example obligations (singleton literal types). Once $\leq$ is defined by the validator's rules, it should validate the expected singleton-to-primitive subtypings for primitive literals (these are representative, actually-implemented cases). (In examples we write `1`, `"x"`, `number`, `string` for the corresponding internal types, by abuse of notation.) For example:

$$\texttt{1} \leq \texttt{number} \qquad \texttt{number} \not\leq \texttt{1}.$$

Similarly, a string literal is a subtype of `string`:

$$\texttt{"x"} \leq \texttt{string}.$$

5 Refinement (Predicate-Driven Narrowing)

5.1 Implemented Refinements

This section enumerates all control-flow refinements implemented in SKJS. When the implementation adds a new refinement, add it to this list.

Master list of implemented refinements.

Category	Test Form	Refines?	Proof
Nullish	`x === null`	Both branches	Lem. 35
	`x === undefined`	Both branches	Lem. 35
`typeof`	`typeof x === "string"`	Both + default	Lem. 36
	`typeof x === "number"`	Both + default	Lem. 36
	`typeof x === "boolean"`	Both + default	Lem. 36
	`typeof x === "bigint"`	Both + default	Lem. 36
	`typeof x === "undefined"`	Both branches	Lem. 36
	`typeof x === "object"`	Both + default	Lem. 36
	`typeof x === "function"`	Both branches	Lem. 36
	`typeof x === "symbol"`	*Not refined*	—
Property	`"p" in x`	Both + default	Lem. 32, 33
Discriminant	`x._tag === ` ℓ	Both branches	Lem. 39
`instanceof`	`x instanceof C`	True only	Lem. 41
Connectives	e_1 `&&` e_2	Compositional	Lem. 29
	e_1 `||` e_2	Compositional	Lem. 29

"Both + default": refines both branches; provides a default type when input is top.

5.2 Soundness Framework

Definition 18 (Refinement Judgements and Soundness)**.** *Let* Γ *map local variables to* TType*; let* σ *map locals to* Val*. Write* $\sigma \models \Gamma$ *when* $\sigma(x) \in [\![\Gamma(x)]\!]$ *for all* x*.*

The judgements $\Gamma \vdash e \overset{\mathsf{t}}{\rightsquigarrow} \Gamma'$ *and* $\Gamma \vdash e \overset{\mathsf{f}}{\rightsquigarrow} \Gamma'$ *refine* Γ *under the assumption that* e *is true/false. The system is* branch-sound *if: whenever* $\sigma \models \Gamma$ *and* e *evaluates to* $b \in \{\mathsf{true}, \mathsf{false}\}$*, then* $\sigma \models \Gamma'$ *for the corresponding derivation.*

Definition 19 (`typeof` Predicates)**.** *Using the* callable *predicate from Definition 7, define the following predicates:*

$$
\begin{aligned}
P_{\mathsf{typeof}=\texttt{"string"}} &\triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = 5\} \\
P_{\mathsf{typeof}=\texttt{"number"}} &\triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = 4\} \\
P_{\mathsf{typeof}=\texttt{"bigint"}} &\triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = 6\} \\
P_{\mathsf{typeof}=\texttt{"boolean"}} &\triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = 3\} \\
P_{\mathsf{typeof}=\texttt{"function"}} &\triangleq \{o \in \mathsf{ObjVal} \mid \mathsf{callable}(o)\} \\
P_{\mathsf{typeof}=\texttt{"undefined"}} &\triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) \in \{0,1\}\} \\
P_{\mathsf{typeof}=\texttt{"object"}} &\triangleq \mathsf{ObjVal} \cup \mathsf{AlgVal} \cup [\![\texttt{null}]\!].
\end{aligned}
$$

Note. *The* `"symbol"` *case is intentionally omitted because SKJS performs no refinement for it.*

Definition 20 (Evaluation for Refinement Tests)**.** *We define a partial evaluation relation for the refinement-test fragment:*

$$\sigma \models_{\mathsf{true}} e \qquad \textit{and} \qquad \sigma \models_{\mathsf{false}} e.$$

It is defined only for the supported test forms:

- ***Nullish:*** $\sigma \models_{\mathsf{true}} (x \texttt{ === } \texttt{null})$ *iff* $\mathsf{rtag}(\sigma(x)) = 2$*;* $\sigma \models_{\mathsf{false}}$ *is the negation. Similarly for* `undefined`*, with* $\mathsf{rtag}(\sigma(x)) \in \{0,1\}$*.*
- ***`typeof`:*** $\sigma \models_{\mathsf{true}} (\texttt{typeof } x \texttt{ === } k)$ *iff* $\sigma(x) \in P_{\mathsf{typeof}=k}$ *where* $P_{\mathsf{typeof}=k}$ *is as in Definition 19; the false relation is negation.*
- ***`in`:*** $\sigma \models_{\mathsf{true}} (\texttt{"}p\texttt{"} \texttt{ in } x)$ *iff* $\sigma(x) \in \mathsf{ObjVal}$ *and* $\mathsf{has}(\sigma(x), p)$*; the false relation is negation.*
- ***Algebraic tag equality:*** $\sigma \models_{\mathsf{true}} (x._\texttt{tag} \texttt{ === } \ell)$ *iff* $\sigma(x) \in \mathsf{AlgVal}$ *and* $\mathsf{atag}(\sigma(x)) = \ell$*, where* $\ell \in \mathsf{String}$ *is a string literal; the false relation is negation.*
- ***`instanceof`:*** $\sigma \models_{\mathsf{true}} (x \texttt{ instanceof } C)$ *iff* $\sigma(x) \in P_{\mathsf{inst}}(C)$ *for a predicate* $P_{\mathsf{inst}}(C) \subseteq \mathsf{Val}$ *(see Definition 40); the false relation is negation.*
- ***Connectives:*** $\sigma \models_{\mathsf{true}} (e_1 \texttt{ \&\& } e_2)$ *iff* $\sigma \models_{\mathsf{true}} e_1$ *and* $\sigma \models_{\mathsf{true}} e_2$*.* $\sigma \models_{\mathsf{false}} (e_1 \texttt{ || } e_2)$ *iff* $\sigma \models_{\mathsf{false}} e_1$ *and* $\sigma \models_{\mathsf{false}} e_2$*. (Other combinations are omitted, since SKJS only uses these connective cases for refinement in the current implementation.)*

Definition 21 (Filter/Map Refinement (Implementation Model))**.** *Let* $P \in \mathcal{P}(\mathsf{Val})$ *be a predicate. Let* $f : \mathsf{SType} \to \{\mathsf{true}, \mathsf{false}\}$ *be a filter and* $g : \mathsf{SType} \to \mathsf{SType}$ *be a mapper. We say* (f, g) *is* sound for P *if for every* $s \in \mathsf{SType}$ *and every* $v \in [\![s]\!]$*:*

$$v \in P \implies (f(s) = \mathsf{true} \ \wedge \ v \in [\![g(s)]\!]).$$

Remark (tag preservation). *In the implementation, the mapper g used for refinement preserves runtime tags:* $\mathsf{tag}(g(s)) = \mathsf{tag}(s)$. *In this document we assume tag preservation when defining* $\mathsf{refine}^{f,g}_{\mathsf{STypes}}$, *so that the result is again a well-formed* STypes *set (pairwise distinct tags). This matches SKJS's* `FilterType` *implementation.*

Lift to STypes. *Define:*

$$\mathsf{refine}^{f,g}_{\mathsf{STypes}}(S) \triangleq \{\, g(s) \mid s \in S,\ f(s) = \mathsf{true} \,\}.$$

Operationally, this is exactly the "filter-map" over the unique representative per tag.

Lift to Clause. *For a clause $c = \langle V \mid S \rangle$ define:*

$$\mathsf{refine}^{f,g}_{\mathsf{Clause}}(\langle V \mid S \rangle) \triangleq \begin{cases} \langle V \mid \top \rangle & \text{if } S = \top \\ \langle V \mid \mathsf{refine}^{f,g}_{\mathsf{STypes}}(S) \rangle & \text{otherwise} \end{cases}$$

If the result is $\langle V \mid \emptyset \rangle$, treat it as $\bot$ (dropped).

Lift to Clauses and TType. *Apply* $\mathsf{refine}^{f,g}_{\mathsf{Clause}}$ *to each clause, dropping $\bot$ results. For $t = (I, C)$:* $\mathsf{refine}^{f,g}(t) \triangleq (I, \mathsf{refine}^{f,g}_{\mathsf{Clauses}}(C))$.

Definition 22 (Defaulting Refinement (Model of `setLocal`))**.** *Some refinements in SKJS provide a* default *type when the current type is too imprecise to refine (notably when it is top or has no informative alternatives). This is modeled by a* defaulting refinement *operator.*

Let $R : \mathsf{TType} \to \mathsf{TType}$ be a transformer and let d be an optional *default simple type ($d \in \{\mathsf{None}\} \cup \{\mathsf{Some}(s) \mid s \in \mathsf{SType}\}$).*

Singleton constructor. *Write* $\mathsf{single}(s)$ *for the* TType *value consisting of a single clause with* $\mathsf{STypes} = \{s\}$ *and no instances.*

Top/empty test. *We say a* TType *value is* syntactically top *if its clause component is $\langle \emptyset \mid \top \rangle$, and* syntactically empty *if its clause component is empty (bottom).*

Definition. *Define* $\mathsf{def}(R, d) : \mathsf{TType} \to \mathsf{TType}$ *by:*

$$\mathsf{def}(R, d)(t) \triangleq \begin{cases} R(t) & \text{if } d = \mathsf{None} \\ \mathsf{single}(s) & \text{if } d = \mathsf{Some}(s) \text{ and } R(t) \text{ is syntactically top or empty} \\ R(t) & \text{otherwise} \end{cases}$$

Note. *This abstracts the implementation check in* `setLocal` *based on* `getSTypes` *(after expansion): it injects a default when refinement would otherwise leave a top/empty shape.*

Definition 23 (Predicate-Sound Type Transformer)**.** *Let $P \in \mathcal{P}(\mathsf{Val})$ and let $R : \mathsf{TType} \to \mathsf{TType}$. We say R is* predicate-sound for P *if for all $t \in \mathsf{TType}$:*

$$[\![t]\!] \cap P \subseteq [\![R(t)]\!].$$

Lemma 24 (Filter/Map Refinement is Predicate-Sound)**.** *If (f, g) is sound for P, then the transformer $t \mapsto \mathsf{refine}^{f,g}(t)$ is predicate-sound for P (Definition 23).*

Proof. Let $t = (\mathsf{insts}, \mathsf{clauses})$. We prove $[\![t]\!] \cap P \subseteq [\![\mathsf{refine}^{f,g}(t)]\!]$ by cases on membership in $[\![\mathsf{insts}]\!] \cup [\![\mathsf{clauses}]\!]$.

- If $v \in [\![\mathsf{insts}]\!] \cap P$, then $v \in [\![\mathsf{insts}]\!] \subseteq [\![\mathsf{refine}^{f,g}(t)]\!]$ since refinement leaves insts unchanged.
- If $v \in [\![\mathsf{clauses}]\!] \cap P$, then $v \in [\![(\mathsf{vars}, \mathsf{stypes})]\!]$ for some clause $(\mathsf{vars}, \mathsf{stypes})$ in clauses. By clause semantics (Definition 16), $v \in \bigcap_{w \in \mathsf{vars}} [\![w]\!]$ and $v \in [\![\mathsf{stypes}]\!]$. If $\mathsf{stypes} = \mathsf{None}$, then $\mathsf{refine}^{f,g}_{\mathsf{Clause}}(\mathsf{vars}, \mathsf{None}) = (\mathsf{vars}, \mathsf{None})$ and the clause remains top, so v is still included. If $\mathsf{stypes} = \mathsf{Some}(S)$, then $v \in [\![S]\!] = \bigcup_{s \in S} [\![s]\!]$, so $v \in [\![s]\!]$ for some $s \in S$. By soundness of (f, g) for P and $v \in P$, we have $f(s) = \mathsf{true}$ and $v \in [\![g(s)]\!]$. Hence $v \in [\![\mathsf{refine}^{f,g}_{\mathsf{STypes}}(S)]\!]$. The refined clause is $(\mathsf{vars}, \mathsf{Some}(\mathsf{refine}^{f,g}_{\mathsf{STypes}}(S)))$, and since vars is unchanged and $v \in \bigcap_{w \in \mathsf{vars}} [\![w]\!]$, we have $v \in [\![(\mathsf{vars}, \mathsf{Some}(\mathsf{refine}^{f,g}_{\mathsf{STypes}}(S)))]\!]$. Therefore $v \in [\![\mathsf{refine}^{f,g}_{\mathsf{Clauses}}(\mathsf{clauses})]\!]$.

Combining cases yields the desired inclusion. □

Lemma 25 (Defaulting Preserves Predicate-Soundness)**.** *Let $R : \mathsf{TType} \to \mathsf{TType}$ be predicate-sound for $P \subseteq \mathsf{Val}$. Let d be an optional default. If $d = \mathsf{None}$, then $\mathsf{def}(R, d)$ is predicate-sound for P. If $d = \mathsf{Some}(s)$ and $P \subseteq [\![\mathsf{single}(s)]\!]$, then $\mathsf{def}(R, d)$ is predicate-sound for P.*

Proof. If $d = \mathsf{None}$, then $\mathsf{def}(R, d) = R$ by Definition 22, so predicate-soundness is immediate. If $d = \mathsf{Some}(s)$, let $t \in \mathsf{TType}$ and let $v \in [\![t]\!] \cap P$. If $R(t)$ is not syntactically top/empty, then $\mathsf{def}(R, d)(t) = R(t)$ and $v \in [\![R(t)]\!]$ by predicate-soundness of R. If $R(t)$ is syntactically top/empty, then $\mathsf{def}(R, d)(t) = \mathsf{single}(s)$, and by hypothesis $v \in P \subseteq [\![\mathsf{single}(s)]\!]$. Thus $[\![t]\!] \cap P \subseteq [\![\mathsf{def}(R, d)(t)]\!]$ for all t. □

5.3 Inference Rules

Filter/Mapper summary. Each test form uses filter f, mapper g, optional default d. True branch: $\mathsf{def}(\mathsf{refine}^{f,g}, d)$; false branch: $\mathsf{refine}^{f',g'}$.

Test	Filter (true)	Mapper	Default
`x === null`	$\mathsf{tag}(s) = 2$	identity	—
`x === undefined`	$\mathsf{tag}(s) \in \{0, 1\}$	identity	—
`typeof x === "string"`	$\mathsf{tag}(s) = 5$	identity	`string`
`typeof x === "number"`	$\mathsf{tag}(s) = 4$	identity	`number`
`typeof x === "function"`	callable object	$\mathsf{Maybe} \to \mathsf{Definite}$	—
`typeof x === "object"`	$\mathsf{tag}(s) \in \{2, 9, 10\}$	identity	`object`$(\emptyset)$
`"p" in x`	tags 7–10	add required p	`object`$(\{p\})$
`x._tag === ℓ`	always true	filter to $\{\ell\}$	—
`x instanceof C`	(overwrite)	$\to T_{\mathsf{inst}}(C)$	—

Local update. $\mathsf{upd}(\Gamma, x, R) \triangleq \Gamma[x \mapsto R(\Gamma(x))]$. If no rule applies, refinement leaves Γ unchanged. Strict inequality `!==` swaps true/false branches.

Nullish filters and mappers.

Definition 26 (Nullish Filters and Mappers)**.** *Define the following filter/map functions on simple types:*

$$g_{\mathrm{id}}(s) \triangleq s.$$

$$f_{\mathsf{null}}(s) \triangleq (\mathsf{tag}(s) = 2), \qquad f_{\neg\mathsf{null}}(s) \triangleq (\mathsf{tag}(s) \neq 2).$$

$$f_{\mathsf{undef}}(s) \triangleq (\mathsf{tag}(s) \in \{0,1\}), \qquad f_{\neg\mathsf{undef}}(s) \triangleq (\mathsf{tag}(s) \notin \{0,1\}).$$

Nullish equality.

$$\frac{}{\Gamma \vdash (x \texttt{ === null}) \overset{\mathsf{t}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\, t \mapsto \mathsf{refine}^{f_{\mathsf{null}}, g_{\mathrm{id}}}(t))}\ \text{R-NullT}$$

$$\frac{}{\Gamma \vdash (x \texttt{ === null}) \overset{\mathsf{f}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\, t \mapsto \mathsf{refine}^{f_{\neg\mathsf{null}}, g_{\mathrm{id}}}(t))}\ \text{R-NullF}$$

$$\frac{}{\Gamma \vdash (x \texttt{ === undefined}) \overset{\mathsf{t}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\, t \mapsto \mathsf{refine}^{f_{\mathsf{undef}}, g_{\mathrm{id}}}(t))}\ \text{R-UndefT}$$

$$\frac{}{\Gamma \vdash (x \texttt{ === undefined}) \overset{\mathsf{f}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\, t \mapsto \mathsf{refine}^{f_{\neg\mathsf{undef}}, g_{\mathrm{id}}}(t))}\ \text{R-UndefF}$$

where $f_{\mathsf{null}}, f_{\neg\mathsf{null}}, f_{\mathsf{undef}}, f_{\neg\mathsf{undef}}, g_{\mathrm{id}}$ are as in Definition 26.

Side condition (locals only). In the implementation, comparisons against `null`/`undefined` are only permitted when the compared expression is a local variable; otherwise a typing error is raised. Accordingly, the rules above are intended only for tests where x is a local.

`typeof` filters and defaults.

Definition 27 (`typeof` Filters and Defaults)**.** *For each k in Definition 19, we define a filter/map pair and an optional default d_k matching the implementation.*

- *For $k \in \{$`"string"`, `"number"`, `"bigint"`, `"boolean"`, `"undefined"`$\}$: $f_k(s)$ is true iff $\mathsf{tag}(s)$ is the corresponding tag, and $g_k \triangleq g_{\mathrm{id}}$.*
- *For $k =$ `"object"`: $f_{\texttt{"object"}}(s)$ is true iff $\mathsf{tag}(s) \in \{2, 9, 10\}$ (object, null, or algebraic), and $g_{\texttt{"object"}} \triangleq g_{\mathrm{id}}$.*
- *For $k =$ `"function"`:*
  - *$f_{\texttt{"function"}}(s) = \mathsf{true}$ iff $s = \texttt{object}(F)$ with $F_{\mathrm{call}} \in \{\mathsf{Definite}(\kappa), \mathsf{Maybe}(\kappa)\}$.*
  - *$g_{\texttt{"function"}}(s)$: if $F_{\mathrm{call}} = \mathsf{Maybe}(\kappa)$, return $\texttt{object}(F')$ with $F'_{\mathrm{call}} = \mathsf{Definite}(\kappa)$; otherwise return s.*

False-branch filter and mapper for `"function"`. *For the false branch of `typeof x === "function"`, we use:*

- *$f_{\neg\texttt{"function"}}(s) = \mathsf{true}$ iff s is not an object, or $s = \texttt{object}(F)$ with $F_{\mathrm{call}} \in \{\mathsf{None}, \mathsf{Maybe}(\kappa)\}$.*
- *$g_{\neg\texttt{"function"}}(s)$: if $F_{\mathrm{call}} = \mathsf{Maybe}(\kappa)$, set $F'_{\mathrm{call}} = \mathsf{None}$; else return s.*

Note. *The key insight is that* $\mathsf{Maybe}(\kappa)$ *types are kept in* both *branches: the true branch refines them to* $\mathsf{Definite}(\kappa)$*, and the false branch refines them to* None*. This ensures predicate-soundness for both branches after normalization. Define defaults:*

- $d_{\texttt{"string"}} = \mathsf{Some}(\texttt{string})$, $d_{\texttt{"number"}} = \mathsf{Some}(\texttt{number})$, $d_{\texttt{"bigint"}} = \mathsf{Some}(\texttt{bigint})$, $d_{\texttt{"boolean"}} = \mathsf{Some}(\texttt{boolean})$.
- *Let* $F_\emptyset$ *be the empty object field specification (no required/optional fields, no index signature, no* `new`/`call` *signature), and set* $d_{\texttt{"object"}} = \mathsf{Some}(\texttt{object}(F_\emptyset))$.
- $d_{\texttt{"function"}} = \mathsf{None}$ *and* $d_{\texttt{"undefined"}} = \mathsf{None}$.

Symbol. *For* $k = \texttt{"symbol"}$*, SKJS performs no refinement, so we omit* f_k, g_k, d_k.

`typeof`.

$$\frac{}{\Gamma \vdash (\texttt{typeof}\ x === k) \overset{\mathsf{t}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\ t \mapsto \mathsf{def}(t' \mapsto \mathsf{refine}^{f_k, g_k}(t'),\ d_k)(t))}\ \text{R-TypeofT}$$

$$\frac{}{\Gamma \vdash (\texttt{typeof}\ x === k) \overset{\mathsf{f}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\ t \mapsto \mathsf{refine}^{f_k^\neg, g_k^\neg}(t))}\ \text{R-TypeofF}$$

where f_k, g_k, d_k are as in Definition 27, and $(f_k^\neg, g_k^\neg) = (\neg f_k, g_{\mathrm{id}})$ for most k, except for $k = \texttt{"function"}$ where $(f^\neg_{\texttt{"function"}}, g^\neg_{\texttt{"function"}}) = (f_{\neg\texttt{"function"}}, g_{\neg\texttt{"function"}})$ as defined in Definition 27.

`typeof "symbol"`. The implementation performs no refinement for `typeof x === "symbol"`; both branches leave the environment unchanged.

`in`.

$$\frac{}{\Gamma \vdash (\texttt{"}p\texttt{"}\ \texttt{in}\ x) \overset{\mathsf{t}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\ t \mapsto \mathsf{def}(t' \mapsto \mathsf{refine}^{f_p^{\mathsf{true}}, g_p^{\mathsf{true}}}(t'),\ d_p^{\mathsf{in}})(t))}\ \text{R-InT}$$

$$\frac{}{\Gamma \vdash (\texttt{"}p\texttt{"}\ \texttt{in}\ x) \overset{\mathsf{f}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\ t \mapsto \mathsf{refine}^{f_p^{\mathsf{false}}, g_p^{\mathsf{false}}}(t))}\ \text{R-InF}$$

where $d_p^{\mathsf{in}} = \mathsf{Some}(\texttt{object}(F_p))$ is a default object type requiring property p with top value type (Definition 7). This corresponds to the implementation's behavior when the left-hand side is a single string literal.

Algebraic tag equality.

$$\frac{}{\Gamma \vdash (x.\texttt{_tag} === \ell) \overset{\mathsf{t}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\ t \mapsto \mathsf{refine}^{f_\ell^{\mathsf{alg}}, g_\ell^{\mathsf{alg,true}}}(t))}\ \text{R-AlgTagT}$$

$$\frac{}{\Gamma \vdash (x.\texttt{_tag} === \ell) \overset{\mathsf{f}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\ t \mapsto \mathsf{refine}^{f_\ell^{\mathsf{alg}}, g_\ell^{\mathsf{alg,false}}}(t))}\ \text{R-AlgTagF}$$

where $\ell \in \mathsf{String}$ is a string literal.

`instanceof`.

$$\frac{}{\Gamma \vdash (x \texttt{ instanceof } C) \overset{\mathsf{t}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\, t \mapsto T_{\mathsf{inst}}(C))}\ \text{R-InstT}$$

$$\frac{}{\Gamma \vdash (x \texttt{ instanceof } C) \overset{\mathsf{f}}{\rightsquigarrow} \Gamma}\ \text{R-InstF}$$

where $T_{\mathsf{inst}}(C)$ is the instance type produced by the implementation (Definition 40).

$$\frac{}{\Gamma \vdash (x \texttt{ instanceof Array}) \overset{\mathsf{t}}{\rightsquigarrow} \mathsf{upd}(\Gamma, x,\, t \mapsto T_{\mathsf{array}})}\ \text{R-InstArrayT}$$

where T_{array} is the implementation's chosen array instance type (e.g. $\texttt{Array}\langle\top\rangle$).

Boolean connectives. The implementation composes refinements sequentially. For example:

$$\frac{\Gamma \vdash e_1 \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_1 \qquad \Gamma_1 \vdash e_2 \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_2}{\Gamma \vdash (e_1 \mathbin{\&\&} e_2) \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_2}\ \text{R-AndT} \qquad \frac{\Gamma \vdash e_1 \overset{\mathsf{f}}{\rightsquigarrow} \Gamma_1 \qquad \Gamma_1 \vdash e_2 \overset{\mathsf{f}}{\rightsquigarrow} \Gamma_2}{\Gamma \vdash (e_1 \mathbin{||} e_2) \overset{\mathsf{f}}{\rightsquigarrow} \Gamma_2}\ \text{R-OrF}$$

Lemma 28 (Predicate-Sound Transformers Yield Branch-Sound Rules)**.** *Let $R : \mathsf{TType} \to \mathsf{TType}$ be predicate-sound for predicate $P \subseteq \mathsf{Val}$. Suppose the inference rule for expression e updates the environment via $\mathsf{upd}(\Gamma, x, R)$. If $\sigma \models_{\mathsf{true}} e$ implies $\sigma(x) \in P$, then the rule is branch-sound for the true branch. Analogously for the false branch with $\sigma \models_{\mathsf{false}} e$.*

Proof. Suppose $\sigma \models \Gamma$ and $\sigma \models_{\mathsf{true}} e$. Let $\Gamma' = \mathsf{upd}(\Gamma, x, R)$. For any local $y \neq x$, $\Gamma'(y) = \Gamma(y)$ and $\sigma(y) \in [\![\Gamma(y)]\!] = [\![\Gamma'(y)]\!]$. For x: by hypothesis $\sigma(x) \in P$, and by $\sigma \models \Gamma$ we have $\sigma(x) \in [\![\Gamma(x)]\!]$. Thus $\sigma(x) \in [\![\Gamma(x)]\!] \cap P \subseteq [\![R(\Gamma(x))]\!] = [\![\Gamma'(x)]\!]$ by predicate-soundness of R. Hence $\sigma \models \Gamma'$. □

Lemma 29 (Connective Rules are Branch-Sound)**.** *Assume that for every derivation $\Gamma \vdash e_1 \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_1$, the rule is branch-sound for the true branch in the sense of Definition 18. Assume likewise that for every derivation $\Gamma_1 \vdash e_2 \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_2$, the rule is branch-sound for the true branch. Then the connective rule **R-AndT** is branch-sound for the true branch.*

*Dually, assume that for every derivation $\Gamma \vdash e_1 \overset{\mathsf{f}}{\rightsquigarrow} \Gamma_1$ and $\Gamma_1 \vdash e_2 \overset{\mathsf{f}}{\rightsquigarrow} \Gamma_2$, the rules are branch-sound for the false branch. Then the connective rule **R-OrF** is branch-sound for the false branch.*

Proof. **R-AndT.** Suppose $\sigma \models \Gamma$ and $\sigma \models_{\mathsf{true}} (e_1 \mathbin{\&\&} e_2)$. By Definition 20, we have $\sigma \models_{\mathsf{true}} e_1$ and $\sigma \models_{\mathsf{true}} e_2$. Consider a derivation of $\Gamma \vdash (e_1 \mathbin{\&\&} e_2) \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_2$ via **R-AndT**; it contains premises $\Gamma \vdash e_1 \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_1$ and $\Gamma_1 \vdash e_2 \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_2$. By branch-soundness of the first premise and $\sigma \models_{\mathsf{true}} e_1$, from $\sigma \models \Gamma$ we obtain $\sigma \models \Gamma_1$. Then by branch-soundness of the second premise and $\sigma \models_{\mathsf{true}} e_2$, we obtain $\sigma \models \Gamma_2$. This is exactly branch-soundness of **R-AndT**.

R-OrF. Suppose $\sigma \models \Gamma$ and $\sigma \models_{\mathsf{false}} (e_1 \mathbin{||} e_2)$. By Definition 20, we have $\sigma \models_{\mathsf{false}} e_1$ and $\sigma \models_{\mathsf{false}} e_2$. For a derivation of $\Gamma \vdash (e_1 \mathbin{||} e_2) \overset{\mathsf{f}}{\rightsquigarrow} \Gamma_2$ via **R-OrF**, we similarly obtain $\sigma \models \Gamma_1$ from the first premise and then $\sigma \models \Gamma_2$ from the second premise. □

Soundness connection. Each of the case-study lemmas in the following subsections establishes that the filter/map pair (f, g) is sound for the corresponding runtime predicate P, which by Lemma 24 implies predicate-soundness of $\mathsf{refine}^{f,g}$, and by Lemma 28 implies branch-soundness of the inference rule.

5.4 Soundness Proofs

The following subsections prove soundness for each refinement category. Each proof shows that the filter/map pair (f, g) is sound for the runtime predicate P, then applies Lemmas 24 and 28.

5.4.1 Property Check (`in`)

The `"p" in x` refinement adds a required field on the true branch, or removes the field constraint on the false branch.

Definition 30 (`in` Predicates)**.** *Fix a property name $p \in \mathsf{String}$. Define:*

$$P_{\text{in}}^{\mathsf{true}}(p) \triangleq \{\, o \in \mathsf{Val} \mid o \in \mathsf{ObjVal} \wedge \mathsf{has}(o, p) \,\} \qquad P_{\text{in}}^{\mathsf{false}}(p) \triangleq \mathsf{Val} \setminus P_{\text{in}}^{\mathsf{true}}(p).$$

Definition 31 (`in` Filter/Map Pair)**.** *Fix $p \in \mathsf{String}$ and let $\top_{\mathsf{TType}}$ be any type with $[\![\top_{\mathsf{TType}}]\!] = \mathsf{Val}$ (e.g. the unique top clause). Let F_p be the object field specification with F_{req} mapping $p \mapsto \top_{\mathsf{TType}}$, F_{opt} empty, $F_{\text{idx}} = \mathsf{None}$, and $F_{\text{new}} = F_{\text{call}} = \mathsf{None}$. Define a mapper $g_p^{\mathsf{true}} : \mathsf{SType} \to \mathsf{SType}$ as follows:*

- *If s is an object type $s = \texttt{object}(F)$ and $F_{\text{idx}} = \mathsf{None}$ and $p \notin \mathrm{dom}(F_{\text{req}})$, then $g_p^{\mathsf{true}}(s)$ is $\texttt{object}(F')$ where F' is F except that F'_{req} additionally maps $p \mapsto \top_{\mathsf{TType}}$.*
- *Otherwise, $g_p^{\mathsf{true}}(s) = s$.*

Let $f_p^{\mathsf{true}} : \mathsf{SType} \to \{\mathsf{true}, \mathsf{false}\}$ be the filter defined by:

$$f_p^{\mathsf{true}}(s) = \begin{cases} \mathsf{true} & \text{if } \mathsf{tag}(s) \in \{7, 8, 9, 10\} \text{ (tuple, promise, object, or algebraic)} \\ \mathsf{false} & \text{otherwise (primitives)} \end{cases}$$

Note. *The implementation conservatively retains tuples, promises, and algebraic types (tags 7, 8, 10) in addition to objects; these cases are marked `TODO` in the implementation. Only object-tagged values (tag 9) can actually satisfy the `in` predicate at runtime, so the filter is sound but imprecise for tags 7, 8, and 10.*

Define a mapper $g_p^{\mathsf{false}} : \mathsf{SType} \to \mathsf{SType}$ by:

- *If $s = \texttt{object}(F)$ then $g_p^{\mathsf{false}}(s) = \texttt{object}(F \setminus p)$ where $F \setminus p$ removes p from both F_{req} and F_{opt} (and leaves $F_{\text{idx}}, F_{\text{new}}, F_{\text{call}}$ unchanged).*
- *Otherwise $g_p^{\mathsf{false}}(s) = s$.*

Let $f_p^{\mathsf{false}}(s) = \mathsf{true}$ for all s (the implementation does not attempt to exclude non-objects in the false branch).

Default. *Define $d_p^{\mathsf{in}} \triangleq \mathsf{Some}(\texttt{object}(F_p))$.*

Lemma 32 (`in`-True Branch (with Default) is Predicate-Sound)**.** *For every $p \in \mathsf{String}$, the transformer $t \mapsto \mathsf{def}(t' \mapsto \mathsf{refine}^{f_p^{\mathsf{true}}, g_p^{\mathsf{true}}}(t'), d_p^{\mathsf{in}})(t)$ is predicate-sound for $P_{\text{in}}^{\mathsf{true}}(p)$.*

Proof. Let $R_p(t) \triangleq \mathsf{refine}^{f_p^{\mathsf{true}}, g_p^{\mathsf{true}}}(t)$. We first show that $(f_p^{\mathsf{true}}, g_p^{\mathsf{true}})$ is sound for $P_{\text{in}}^{\mathsf{true}}(p)$ (Definition 21), and conclude that R_p is predicate-sound for $P_{\text{in}}^{\mathsf{true}}(p)$ by Lemma 24. Let $s \in \mathsf{SType}$ and $v \in [\![s]\!]$ with $v \in P_{\text{in}}^{\mathsf{true}}(p)$. Then $v \in \mathsf{ObjVal}$, so $\mathsf{rtag}(v) = 9$. By A0, $\mathsf{rtag}(v) = \mathsf{tag}(s)$, hence $\mathsf{tag}(s) = 9$ and therefore s must be an object type of the form $\texttt{object}(F)$. By definition of $P_{\text{in}}^{\mathsf{true}}(p)$,

$\mathsf{has}(v, p)$. The filter keeps object types, so $f_p^{\mathsf{true}}(s) = \mathsf{true}$. If g_p^{true} adds the required field $p \mapsto \top_{\mathsf{TType}}$, then v satisfies the added requirement because it already has p and $\mathsf{get}(v, p) \in [\![\top_{\mathsf{TType}}]\!] = \mathsf{Val}$. Otherwise $g_p^{\mathsf{true}}(s) = s$ and membership is immediate. Thus $v \in [\![g_p^{\mathsf{true}}(s)]\!]$ and soundness holds.

Finally, by Definition 7, every $v \in P_{\text{in}}^{\mathsf{true}}(p)$ is in $[\![\texttt{object}(F_p)]\!] = [\![\mathsf{single}(\texttt{object}(F_p))]\!]$, so $P_{\text{in}}^{\mathsf{true}}(p) \subseteq [\![\mathsf{single}(\texttt{object}(F_p))]\!]$. By Lemma 25, $\mathsf{def}(R_p, d_p^{\mathsf{in}})$ is predicate-sound for $P_{\text{in}}^{\mathsf{true}}(p)$. □

Lemma 33 (`in`-False Branch is Predicate-Sound)**.** *For every $p \in \mathsf{String}$, the refinement $\mathsf{refine}^{f_p^{\mathsf{false}}, g_p^{\mathsf{false}}}$ is predicate-sound for $P_{\text{in}}^{\mathsf{false}}(p)$.*

Proof. We show $(f_p^{\mathsf{false}}, g_p^{\mathsf{false}})$ is sound for $P_{\text{in}}^{\mathsf{false}}(p)$ (Definition 21), and conclude by Lemma 24. Let $s \in \mathsf{SType}$ and $v \in [\![s]\!]$ with $v \in P_{\text{in}}^{\mathsf{false}}(p)$. We have $f_p^{\mathsf{false}}(s) = \mathsf{true}$ by definition. If s is not an object type, then $g_p^{\mathsf{false}}(s) = s$ so $v \in [\![g_p^{\mathsf{false}}(s)]\!]$. If $s = \texttt{object}(F)$, then $g_p^{\mathsf{false}}(s) = \texttt{object}(F \setminus p)$ which removes p from field constraints. By Definition 7, the semantics only constrains listed fields, so removing a field enlarges (or preserves) the inhabitant set: $[\![\texttt{object}(F)]\!] \subseteq [\![\texttt{object}(F \setminus p)]\!]$. Hence $v \in [\![g_p^{\mathsf{false}}(s)]\!]$. □

5.4.2 Nullish Checks

Definition 34 (Nullish Predicates)**.** *Define the following predicates on runtime values:*

$$P_{\mathsf{null}} \triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = 2\} \qquad P_{\mathsf{undef}} \triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) \in \{0, 1\}\}.$$

We also write $\neg P$ for $\mathsf{Val} \setminus P$.

Lemma 35 (Nullish Refinement is Predicate-Sound)**.** *The refinements for null and undefined (Definition 26) are predicate-sound for their respective predicates.*

Proof. By A0: if $v \in P_{\mathsf{null}}$ then $\mathsf{rtag}(v) = 2$, so $\mathsf{tag}(s) = 2$ and $f_{\mathsf{null}}(s) = \mathsf{true}$. Apply Lemma 24. The undefined and negated cases are analogous. □

5.4.3 `typeof` Checks

Predicates are in Definition 19; filters/defaults in Definition 27.

Lemma 36 (`typeof` Refinement is Predicate-Sound)**.** *For each k in Definition 19, the transformer $t \mapsto \mathsf{refine}^{f_k, g_k}(t)$ is predicate-sound for $P_{\mathsf{typeof}=k}$, and the false-branch transformer $t \mapsto \mathsf{refine}^{f_k^\neg, g_k^\neg}(t)$ is predicate-sound for $\neg P_{\mathsf{typeof}=k}$, where $(f_k^\neg, g_k^\neg) = (\neg f_k, g_{\text{id}})$ for most k, except for $k =$ `"function"` where $(f^\neg_{\texttt{"function"}}, g^\neg_{\texttt{"function"}}) = (f_{\neg\texttt{"function"}}, g_{\neg\texttt{"function"}})$ as in Definition 27.*

Defaults. *If $d_k = \mathsf{Some}(s_k)$ satisfies $P_{\mathsf{typeof}=k} \subseteq [\![\mathsf{single}(s_k)]\!]$, then the defaulted transformer*

$$t \mapsto \mathsf{def}(t' \mapsto \mathsf{refine}^{f_k, g_k}(t'), d_k)(t)$$

is predicate-sound for $P_{\mathsf{typeof}=k}$.

Proof. For tag-based cases ($k \in \{\texttt{"string"}, \texttt{"number"}, \ldots\}$): by A0, $v \in P_{\mathsf{typeof}=k}$ implies $\mathsf{tag}(s)$ matches the predicate's tag-set, so $f_k(s) = \mathsf{true}$. Since $g_k = g_{\text{id}}$, apply Lemma 24.

For $k =$ `"function"`: the key is that $\mathsf{callable}(v) = \mathsf{true}$ forces $F_{\text{call}} \neq \mathsf{None}$, so the mapper upgrading $\mathsf{Maybe}(\kappa) \to \mathsf{Definite}(\kappa)$ preserves v.

For the false branch of `"function"`: $\mathsf{callable}(v) = \mathsf{false}$ implies $F_{\text{call}} \in \{\mathsf{None}, \mathsf{Maybe}(\kappa)\}$, and the mapper downgrading to None preserves v.

Defaults follow from Lemma 25. □

5.4.4 Discriminant (_tag) Checks

Refines algebraic types by filtering the constructor tag set.

Definition 37 (Algebraic-Tag Predicate)**.** *Fix $\ell \in \mathsf{String}$ (a tag label) and define:*

$$P_{\mathsf{algTag}=\ell} \triangleq \{v \in \mathsf{Val} \mid v \in \mathsf{AlgVal} \wedge \mathsf{atag}(v) = \ell\}.$$

Definition 38 (Algebraic Tag Filter/Map Pairs)**.** *Fix $\ell \in \mathsf{String}$. Let $f_\ell^{\mathsf{alg}}(s) = \mathsf{true}$ for all s (the implementation does not drop non-algebraic alternatives for this refinement). Define mappers:*

- $g_\ell^{\mathsf{alg,true}}(\texttt{algebraic}(T)) \triangleq \texttt{algebraic}(T \cap \{\ell\})$*, and* $g_\ell^{\mathsf{alg,true}}(s) \triangleq s$ *for all other s.*
- $g_\ell^{\mathsf{alg,false}}(\texttt{algebraic}(T)) \triangleq \texttt{algebraic}(T \setminus \{\ell\})$*, and* $g_\ell^{\mathsf{alg,false}}(s) \triangleq s$ *for all other s.*

Lemma 39 (Algebraic Tag Equality is Predicate-Sound)**.** *For every $\ell \in \mathsf{String}$:*

$$\mathsf{refine}^{f_\ell^{\mathsf{alg}}, g_\ell^{\mathsf{alg,true}}} \text{ is predicate-sound for } P_{\mathsf{algTag}=\ell},$$
$$\mathsf{refine}^{f_\ell^{\mathsf{alg}}, g_\ell^{\mathsf{alg,false}}} \text{ is predicate-sound for } \neg P_{\mathsf{algTag}=\ell}.$$

Proof. True branch: If $v \in P_{\mathsf{algTag}=\ell}$ then $\mathsf{atag}(v) = \ell$, so $\ell \in T$ for $s = \texttt{algebraic}(T)$, and the mapper restricting to $\{ell\}$ preserves v.

False branch: If $\mathsf{atag}(v) \neq \ell$, then $\mathsf{atag}(v) \in T \setminus \{\ell\}$, so the mapper removing ℓ preserves v. Non-algebraic types pass through unchanged. □

5.4.5 `instanceof` Checks

Refines the true branch by overwriting with the instance type $T_{\mathsf{inst}}(C)$.

Definition 40 (`instanceof` Semantics and Implementation-Modeled Refinement)**.** *Assume a set of runtime class names* Class *and a predicate family*

$$P_{\mathsf{inst}} : \mathsf{Class} \to \mathcal{P}(\mathsf{Val})$$

capturing the runtime truth of `instanceof` *for each class. We assume the type system's instance forms in* TType *include a corresponding instance type $T_{\mathsf{inst}}(C)$ (with implementation-chosen type arguments), and a distinguished array instance type T_{array}.*

Axiom (`instanceof` inclusion). *For each class C we assume:*

$$v \in P_{\mathsf{inst}}(C) \implies v \in [\![T_{\mathsf{inst}}(C)]\!].$$

Array special case. *We assume $P_{\mathsf{inst}}(\texttt{Array}) \subseteq [\![T_{\mathsf{array}}]\!]$.*

Lemma 41 (`instanceof`-True Overwrite is Predicate-Sound)**.** *For each class C, the constant transformer $R_C(t) \triangleq T_{\mathsf{inst}}(C)$ is predicate-sound for $P_{\mathsf{inst}}(C)$. Similarly, the constant transformer $R_{\mathsf{array}}(t) \triangleq T_{\mathsf{array}}$ is predicate-sound for $P_{\mathsf{inst}}(\texttt{Array})$.*

Proof. Fix C and let $t \in \mathsf{TType}$. If $v \in [\![t]\!] \cap P_{\mathsf{inst}}(C)$, then $v \in P_{\mathsf{inst}}(C)$. By the axiom in Definition 40, $v \in [\![T_{\mathsf{inst}}(C)]\!] = [\![R_C(t)]\!]$. Thus $[\![t]\!] \cap P_{\mathsf{inst}}(C) \subseteq [\![R_C(t)]\!]$. The array case is identical using $P_{\mathsf{inst}}(\texttt{Array}) \subseteq [\![T_{\mathsf{array}}]\!]$. □

6 Type Difference (Diff)

Definition 42 (Diff Soundness Properties)**.** *Let X be a set equipped with a semantic interpretation $[\![\cdot]\!] : X \to \mathcal{P}(\mathsf{Val})$. An operation $\mathsf{diff} : X \times X \to X$ is* sound *(with respect to $[\![\cdot]\!]$) if for all $x_1, x_2 \in X$, letting $x_3 = \mathsf{diff}(x_1, x_2)$, the following properties hold:*

$$\textit{(P1)} \quad [\![x_3]\!] \subseteq [\![x_1]\!] \qquad \text{(under-approximation)}$$
$$\textit{(P2)} \quad [\![x_3]\!] \cup [\![x_2]\!] \supseteq [\![x_1]\!] \qquad \text{(coverage)}$$

In this document we instantiate X with STypes*,* Clause*,* $\mathsf{Clauses}$*, and* TType*.*

Reader's guide. *These two properties imply that $[\![x_3]\!]$ is always a refinement of $[\![x_1]\!]$, and that it still covers all of $[\![x_1]\!]$ once $[\![x_2]\!]$ is added back. Equivalently (Lemma 43), $[\![x_3]\!]$ is an* over-approximation *of the true set difference $[\![x_1]\!] \setminus [\![x_2]\!]$ while staying within $[\![x_1]\!]$.*

Lemma 43 (Equivalent Characterization)**.** *For all $x_1, x_2, x_3 \in X$, (P1) $\wedge$ (P2) holds if and only if*

$$[\![x_1]\!] \setminus [\![x_2]\!] \subseteq [\![x_3]\!] \subseteq [\![x_1]\!].$$

Proof. ($\Rightarrow$) Let $v \in [\![x_1]\!] \setminus [\![x_2]\!]$. By (P2), $v \in [\![x_3]\!] \cup [\![x_2]\!]$. Since $v \notin [\![x_2]\!]$, we have $v \in [\![x_3]\!]$. Thus $[\![x_1]\!] \setminus [\![x_2]\!] \subseteq [\![x_3]\!]$. Combined with (P1): $[\![x_1]\!] \setminus [\![x_2]\!] \subseteq [\![x_3]\!] \subseteq [\![x_1]\!]$.

($\Leftarrow$) The right-hand inclusion $[\![x_3]\!] \subseteq [\![x_1]\!]$ is exactly (P1). For (P2),

$$[\![x_3]\!] \cup [\![x_2]\!] \supseteq ([\![x_1]\!] \setminus [\![x_2]\!]) \cup [\![x_2]\!] \supseteq [\![x_1]\!].$$

□

Definition 44 (Completeness)**.** *A diff operation is* complete *if:*

$$[\![\mathsf{diff}(t_1, t_2)]\!] = [\![t_1]\!] \setminus [\![t_2]\!]$$

By the equivalent characterization, (P1) $\wedge$ (P2) implies $[\![t_1]\!] \setminus [\![t_2]\!] \subseteq [\![\mathsf{diff}(t_1, t_2)]\!]$. Completeness additionally requires the reverse: $[\![\mathsf{diff}(t_1, t_2)]\!] \subseteq [\![t_1]\!] \setminus [\![t_2]\!]$.

6.1 STypes Diff

Definition 45 (STypes Diff)**.**

$$\mathsf{diff}_{\mathsf{STypes}}(L, R) \triangleq \{\, s \in L \mid \mathsf{tag}(s) \notin \mathsf{tags}(R) \,\}.$$

Intuition. *Since an* STypes *value stores at most one representative per runtime tag, this diff only removes whole tags: it preserves exactly the tags present on the left but absent on the right.*

Lemma 46 (STypes Diff Satisfies (P1))**.** $[\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!] \subseteq [\![L]\!]$.

Proof. $\mathsf{diff}_{\mathsf{STypes}}(L, R) \subseteq L$ by definition, so:

$$[\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!] = \bigcup_{s \in \mathsf{diff}_{\mathsf{STypes}}(L,R)} [\![s]\!] \subseteq \bigcup_{s \in L} [\![s]\!] = [\![L]\!].$$

□

Definition 47 (Same-Tag Coverage Condition)**.** *For $L, R \in \mathsf{STypes}$, define:*

$$\mathsf{STC}(L, R) \iff \forall k \in \mathsf{tags}(L) \cap \mathsf{tags}(R).\ [\![L[k]]\!] \subseteq [\![R[k]]\!].$$

Lemma 48 (STypes Diff Satisfies (P2) under STC)**.** *If $\mathsf{STC}(L, R)$ holds, then $[\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!] \cup [\![R]\!] \supseteq [\![L]\!]$.*

Proof. Let $x \in [\![L]\!]$. Then $x \in [\![L[k]]\!]$ for some $k \in \mathsf{tags}(L)$.

- Case $k \notin \mathsf{tags}(R)$: Let $s = L[k]$. Then $\mathsf{tag}(s) = k$ and $s \in \mathsf{diff}_{\mathsf{STypes}}(L, R)$, so $x \in [\![s]\!] \subseteq [\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!]$.
- Case $k \in \mathsf{tags}(R)$: By STC, $[\![L[k]]\!] \subseteq [\![R[k]]\!] \subseteq [\![R]\!]$, so $x \in [\![R]\!]$.

□

Corollary 49 (STypes Diff under STC)**.** *If $\mathsf{STC}(L, R)$ holds, then (P2) holds. In particular, if $L[k] = R[k]$ for all $k \in \mathsf{tags}(L) \cap \mathsf{tags}(R)$, then (P2) holds.*

Lemma 50 (STypes Diff Completeness)**.** $\mathsf{diff}_{\mathsf{STypes}}$ *is complete (i.e., $[\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!] = [\![L]\!] \setminus [\![R]\!]$) if and only if $\mathsf{STC}(L, R)$.*

Proof. ($\Rightarrow$) Suppose completeness holds and let $k \in \mathsf{tags}(L) \cap \mathsf{tags}(R)$. Let $s_L = L[k]$ and $s_R = R[k]$. Since $k \in \mathsf{tags}(R)$, we have $s_L \notin \mathsf{diff}_{\mathsf{STypes}}(L, R)$, so by completeness $[\![s_L]\!] \subseteq [\![R]\!]$. By A1 and the distinct-tag invariant for STypes, $[\![s_L]\!]$ can only overlap with the unique same-tag alternative $[\![s_R]\!]$, hence $[\![s_L]\!] \subseteq [\![s_R]\!]$, i.e. $[\![L[k]]\!] \subseteq [\![R[k]]\!]$.

($\Leftarrow$) We show $[\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!] = [\![L]\!] \setminus [\![R]\!]$.

($\subseteq$) Let $x \in [\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!]$. Then $x \in [\![s]\!]$ for some $s \in \mathsf{diff}_{\mathsf{STypes}}(L, R)$. Let $k = \mathsf{tag}(s)$. Then $k \notin \mathsf{tags}(R)$, so by A1 no element of R has tag k, hence $x \notin [\![R]\!]$. Also $s \in L$, so $x \in [\![L]\!]$. Therefore $x \in [\![L]\!] \setminus [\![R]\!]$.

($\supseteq$) Let $x \in [\![L]\!] \setminus [\![R]\!]$. Then $x \in [\![L[k]]\!]$ for some $k \in \mathsf{tags}(L)$. If $k \in \mathsf{tags}(R)$, then by hypothesis $[\![L[k]]\!] \subseteq [\![R[k]]\!] \subseteq [\![R]\!]$, contradicting $x \notin [\![R]\!]$. So $k \notin \mathsf{tags}(R)$, hence $L[k] \in \mathsf{diff}_{\mathsf{STypes}}(L, R)$ and $x \in [\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!]$. □

Corollary 51 (When STypes Diff is Incomplete)**.** $\mathsf{diff}_{\mathsf{STypes}}(L, R)$ *is incomplete when there exists $k \in \mathsf{tags}(L) \cap \mathsf{tags}(R)$ with $[\![L[k]]\!] \not\subseteq [\![R[k]]\!]$. Example: $\mathsf{diff}_{\mathsf{STypes}}(\{\texttt{number}\}, \{\texttt{1}\}) = \emptyset$, but $[\![\texttt{number}]\!] \setminus [\![\texttt{1}]\!] \neq \emptyset$.*

6.2 TInsts Diff

Definition 52 (TInsts Diff)**.**

$$\mathsf{diff}_{\mathsf{TInsts}}(L, R) = L \setminus R = \{i \in L \mid i \notin R\}$$

Note. *In the implementation,* TInsts *is represented as a collection of instance AST nodes and* $\mathsf{diff}_{\mathsf{TInsts}}$ *removes elements based on* structural equality *of instances. In this paper, that corresponds exactly to the usual mathematical set difference on the set* TInst *(i.e. membership/equality is the syntactic identity of instances, not semantic overlap).*

Lemma 53 (TInsts Diff Satisfies (P1) and (P2))**.** $[\![\mathsf{diff}_{\mathsf{TInsts}}(L, R)]\!] \subseteq [\![L]\!]$ *and* $[\![\mathsf{diff}_{\mathsf{TInsts}}(L, R)]\!] \cup [\![R]\!] \supseteq [\![L]\!]$.

Proof. (P1): $\mathsf{diff}_{\mathsf{TInsts}}(L, R) \subseteq L$, so $[\![\mathsf{diff}_{\mathsf{TInsts}}(L, R)]\!] \subseteq [\![L]\!]$.

(P2): Let $x \in [\![L]\!]$. Then $x \in [\![i]\!]$ for some $i \in L$.

- If $i \in R$: Then $x \in [\![i]\!] \subseteq [\![R]\!]$.
- If $i \notin R$: Then $i \in \mathsf{diff}_{\mathsf{TInsts}}(L, R)$, so $x \in [\![\mathsf{diff}_{\mathsf{TInsts}}(L, R)]\!]$.

□

Lemma 54 (TInsts Diff Completeness)**.** $\mathsf{diff}_{\mathsf{TInsts}}$ *is complete if and only if for all* $i \in L \setminus R$ *and* $j \in R$: $[\![i]\!] \cap [\![j]\!] = \emptyset$.

Proof. ($\Rightarrow$) Suppose completeness and let $i \in L \setminus R$ and $j \in R$. If $x \in [\![i]\!] \cap [\![j]\!]$, then $i \in \mathsf{diff}_{\mathsf{TInsts}}(L, R)$, so $x \in [\![\mathsf{diff}_{\mathsf{TInsts}}(L, R)]\!] = [\![L]\!] \setminus [\![R]\!]$. But $x \in [\![j]\!] \subseteq [\![R]\!]$, contradiction.

($\Leftarrow$) We show $[\![\mathsf{diff}_{\mathsf{TInsts}}(L, R)]\!] = [\![L]\!] \setminus [\![R]\!]$.

($\subseteq$) Let $x \in [\![\mathsf{diff}_{\mathsf{TInsts}}(L, R)]\!]$. Then $x \in [\![i]\!]$ for some $i \in L \setminus R$. For any $j \in R$, $[\![i]\!] \cap [\![j]\!] = \emptyset$ by hypothesis. Thus $x \notin [\![R]\!]$, so $x \in [\![L]\!] \setminus [\![R]\!]$.

($\supseteq$) Let $x \in [\![L]\!] \setminus [\![R]\!]$. Then $x \in [\![i]\!]$ for some $i \in L$. If $i \in R$, then $x \in [\![R]\!]$, contradiction. So $i \in L \setminus R = \mathsf{diff}_{\mathsf{TInsts}}(L, R)$. □

6.3 Clause Diff

Definition 55 (Clause Diff)**.** *For clauses* $c_1 = \langle V_1 \mid S_1 \rangle$ *and* $c_2 = \langle V_2 \mid S_2 \rangle$*:*

$$\mathsf{diff}_{\mathsf{Clause}}(c_1, c_2) = \begin{cases} \bot & \text{if } S_2 = \top \\ c_1 & \text{if } S_1 = \top \\ \langle V_1 \mid \mathsf{diff}_{\mathsf{STypes}}(S_1, S_2) \rangle & \text{otherwise} \end{cases}$$

where $\bot$ *(empty type) has* $[\![\bot]\!] = \emptyset$.

Lemma 56 (Clause Diff Satisfies (P1))**.** $[\![\mathsf{diff}_{\mathsf{Clause}}(c_1, c_2)]\!] \subseteq [\![c_1]\!]$.

Proof. By case analysis on S_1, S_2. The non-trivial case uses Lemma 46. □

Lemma 57 (Clause Diff Satisfies (P2) Conditionally)**.** $\mathsf{diff}_{\mathsf{Clause}}$ *satisfies (P2) when:*

1. $S_2 = \top$*: Always (since* $[\![c_2]\!] = \mathsf{Val} \supseteq [\![c_1]\!]$*).*
2. $S_1 = \top$*: Always (result is* c_1*).*
3. *Otherwise: When* $\mathsf{diff}_{\mathsf{STypes}}(S_1, S_2)$ *satisfies (P2).*

6.4 Clauses Diff

Definition 58 (Clauses Diff)**.** *For* $C_1, C_2 \in \mathsf{Clauses}$*, let* $c_\emptyset = C_2(\emptyset)$ *if* $\emptyset \in \mathrm{dom}(C_2)$*, else undefined.*

$$\mathsf{diff}_{\mathsf{Clauses}}(C_1, C_2)(k) = \begin{cases} \mathsf{diff}_{\mathsf{Clause}}(\mathsf{diff}_{\mathsf{Clause}}(C_1(k), C_2(k)), c_\emptyset) & \text{if } k \in \mathrm{dom}(C_2) \wedge c_\emptyset \text{ defined} \\ \mathsf{diff}_{\mathsf{Clause}}(C_1(k), C_2(k)) & \text{if } k \in \mathrm{dom}(C_2) \wedge c_\emptyset \text{ undefined} \\ \mathsf{diff}_{\mathsf{Clause}}(C_1(k), c_\emptyset) & \text{if } k \notin \mathrm{dom}(C_2) \wedge c_\emptyset \text{ defined} \\ C_1(k) & \text{if } k \notin \mathrm{dom}(C_2) \wedge c_\emptyset \text{ undefined} \end{cases}$$

for $k \in \mathrm{dom}(C_1)$*. Clauses mapped to* $\bot$ *are dropped.*

Note. *Whenever $k \in \mathrm{dom}(C_2)$, the call $\mathsf{diff}_{\mathsf{Clause}}(C_1(k), C_2(k))$ compares clauses stored under the* same key *k, so their variable-sets match by the* Clauses *well-formedness condition (keys agree with clause variables). The only subtraction that may use a different key is via $c_\emptyset = C_2(\emptyset)$, which is deliberately applied to every $C_1(k)$; this corresponds to the implementation behavior and explains why the right-hand clause's variables are not used by* $\mathsf{diff}_{\mathsf{Clause}}$.

Why the $\emptyset$-clause matters. *The distinguished key $\emptyset$ corresponds to clauses that do not require any type parameters. When present on the right, subtracting $c_\emptyset$ from every left clause models a "global" elimination that should apply regardless of which type-parameter constraints produced the left clause.*

Lemma 59 (Clauses Diff Satisfies (P1))**.** $[\![\mathsf{diff}_{\mathsf{Clauses}}(C_1, C_2)]\!] \subseteq [\![C_1]\!]$.

Proof. Each output clause is derived from $C_1(k)$ via $\mathsf{diff}_{\mathsf{Clause}}$, which satisfies (P1). □

6.5 TType Diff

Definition 60 (TType Diff)**.** *For $t_1 = (\mathsf{insts}_1, \mathsf{clauses}_1)$ and $t_2 = (\mathsf{insts}_2, \mathsf{clauses}_2)$:*

$$\mathsf{diff}_{\mathsf{TType}}(t_1, t_2) = (\mathsf{diff}_{\mathsf{TInsts}}(\mathsf{insts}_1, \mathsf{insts}_2), \mathsf{diff}_{\mathsf{Clauses}}(\mathsf{clauses}_1, \mathsf{clauses}_2))$$

Theorem 61 (TType Diff Satisfies (P1))**.** $[\![\mathsf{diff}_{\mathsf{TType}}(t_1, t_2)]\!] \subseteq [\![t_1]\!]$.

Proof.

$$\begin{aligned}[\![\mathsf{diff}_{\mathsf{TType}}(t_1, t_2)]\!] &= [\![\mathsf{diff}_{\mathsf{TInsts}}(\mathsf{insts}_1, \mathsf{insts}_2)]\!] \cup [\![\mathsf{diff}_{\mathsf{Clauses}}(\mathsf{clauses}_1, \mathsf{clauses}_2)]\!] \\ &\subseteq [\![\mathsf{insts}_1]\!] \cup [\![\mathsf{clauses}_1]\!] = [\![t_1]\!]\end{aligned}$$

by Lemmas 53 and 59. □

Theorem 62 (TType Diff Satisfies (P2) Conditionally)**.** $\mathsf{diff}_{\mathsf{TType}}$ *satisfies (P2) when all underlying* $\mathsf{diff}_{\mathsf{STypes}}$ *calls satisfy (P2). In particular, (P2) holds when* STC *holds for all same-tag eliminations.*

Theorem 63 (TType Diff Completeness)**.** $\mathsf{diff}_{\mathsf{TType}}$ *is complete when:*

1. $\mathsf{diff}_{\mathsf{TInsts}}$ *is complete (structurally distinct instances are semantically disjoint), and*
2. $\mathsf{diff}_{\mathsf{STypes}}$ *is complete for all clause pairs (same-tag implies $[\![s]\!] \subseteq [\![s']\!]$).*

Corollary 64 (Explicit Completeness Conditions for $\mathsf{diff}_{\mathsf{TType}}$)**.** *Let $t_1 = (\mathsf{insts}_1, \mathsf{clauses}_1)$ and $t_2 = (\mathsf{insts}_2, \mathsf{clauses}_2)$. Then $\mathsf{diff}_{\mathsf{TType}}(t_1, t_2)$ is complete (i.e. $[\![\mathsf{diff}_{\mathsf{TType}}(t_1, t_2)]\!] = [\![t_1]\!] \setminus [\![t_2]\!]$) under the following explicit hypotheses:*

1. ***Instance disjointness:*** *For all $i \in \mathsf{insts}_1$ and $j \in \mathsf{insts}_2$ with $i \neq j$, we have $[\![i]\!] \cap [\![j]\!] = \emptyset$.*
2. ***Top-clause alignment:*** *If $\mathsf{clauses}_1$ contains $\langle \emptyset \mid \top \rangle$, then $\mathsf{clauses}_2$ also contains it.*
3. ***Same-tag coverage for clause subtractions:*** *For each $k \in \mathrm{dom}(\mathsf{clauses}_1) \cap \mathrm{dom}(\mathsf{clauses}_2)$ with $\mathsf{clauses}_i(k) = \langle k \mid S_i \rangle$: $\mathsf{STC}(S_1, S_2)$. Similarly for the empty-key clause if present in $\mathsf{clauses}_2$.*

7 Application to Type Inference

Definition 65 (Subtyping Check with Validation)**.** *When checking $u_1 \leq u_2$ where u_2 contains type variable T:*

1. *Compute* $\mathsf{diff}(u_1, u_2)$ *and grow T's lower bound.*
2. *Expand u_1, u_2 using the substitution.*
3. *Validate via* $\mathsf{isSub}(expand(u_1), expand(u_2))$*.*

Theorem 66 (Inference Soundness)**.** *If* isSub *correctly validates subtyping on expanded types, then type inference is sound regardless of whether* diff *satisfies (P2).*

Proof. The diff result is used only to guide inference (e.g. by accumulating bounds or hints about T). The acceptance/rejection decision is made solely by isSub on expanded types. Therefore, an imprecise diff can at worst cause incompleteness (e.g. by failing to discover a useful bound, leading to a later rejection), but it cannot cause an unsound acceptance when isSub is correct. □

8 Examples of Incompleteness

We present two concrete examples of incompleteness in SKJS:

- a *control-flow refinement* case that TypeScript accepts but SKJS currently fails to narrow (a missing implementation case), and
- a diff*-based* case where diff is incomplete.

8.1 Example 1: Missing Strict-Equality Refinement for Primitive Literals

TypeScript code:

```
function f(x: 1 | 2): 2 {
  if (x === 1) { return 2; }
  return x;  // x should be narrowed to 2
}
```

Mathematical setup: Both `1` and `2` have the same runtime tag 4. SKJS represents primitive literal unions by storing a finite literal-set payload inside a single tag representative, so `1 | 2` is representable and the narrowed type `2` is expressible. Unlike Example 2, this is not a representational limitation.

Incompleteness (missing implementation case): The control-flow refinement for strict equality in SKJS currently handles only:

1. algebraic discriminant comparisons (e.g. `x._tag === "Foo"`),
2. `typeof` comparisons, and
3. comparisons of a local to `null` or `undefined`.

It does *not* implement refinement for comparisons of a local against a primitive literal such as `x === 1`. Therefore, SKJS does not narrow `x` in either branch and rejects `return x;`, even though TypeScript narrows `x` to `2`.

Could it be implemented? Yes. Since `1|2` is stored as a finite literal-set at the numeric tag, the true branch could intersect the payload with {1} and the false branch could subtract {1}, yielding {2}. This requires adding a strict-equality refinement case for primitive literals, not changing the STypes normal form.

8.2 Example 2: Subtracting from Top (`unknown`)

Note on {} syntax. In TypeScript (with `--strictNullChecks`), the type {} behaves as a "non-nullish top": it includes all values except `null` and `undefined`, including primitives such as numbers and strings. In SKJS, the surface syntax {} denotes an *object* type (runtime tag 9) and does *not* include primitives. The example below uses TypeScript's interpretation of {}.

TypeScript code (requires `--strictNullChecks`):

```
function f(x: unknown): {} {
  if (x !== null && x !== undefined) {
    return x;  // x should be narrowed to {}
  }
  return {};
}
```

Mathematical setup: The type `unknown` is represented as $\langle \emptyset \mid \top \rangle$. Let $c_1 = \langle \emptyset \mid \top \rangle$ represent `unknown`, and $c_2 = \langle \emptyset \mid \{\texttt{undefined}, \texttt{null}\} \rangle$ represent `null | undefined`.

Incompleteness: By definition of $\mathsf{diff}_{\mathsf{Clause}}$:

$$\mathsf{diff}_{\mathsf{Clause}}(c_1, c_2) = c_1 \quad (\text{since } S_1 = \top)$$

So $[\![\mathsf{diff}_{\mathsf{Clause}}(c_1, c_2)]\!] = [\![c_1]\!] = \mathsf{Val}$ (all values). But the true set difference is:

$$[\![c_1]\!] \setminus [\![c_2]\!] = \mathsf{Val} \setminus ([\![\texttt{null}]\!] \cup [\![\texttt{undefined}]\!]) \neq \mathsf{Val}$$

The result is incomplete: diff returns `unknown` instead of the narrowed non-nullish type. The clause diff cannot "subtract from top" because None does not enumerate tags—there is no finite set from which to remove the subtracted alternatives. The implementation in `Clause::diff` simply returns the left clause unchanged when $\mathsf{stypes}_1 = \mathsf{None}$.

Could it be implemented? Yes, but it requires a representation extension. To model TypeScript's non-nullish type precisely, one could add a dedicated form (e.g. a *non-nullish* flag, or an *excluded-tag* mechanism) rather than attempting to enumerate "all other tags" inside STypes. The current SKJS implementation chooses neither approach and leaves top unchanged when subtracting a non-top right-hand side.

Making this example work in SKJS. To express this pattern in SKJS, one would need a surface spelling for non-nullish types (such as `NonNullable<T>`) and an internal representation that can express "remove null/undefined" from an arbitrary type, including top.

A Implementation Correspondences

This appendix summarizes the correspondence between the mathematical model and SKJS implementation.

Mathematical Concept	SKJS Implementation
SType (simple type)	`SType` enum with tag-indexed variants
STypes (tag-distinct set)	`STypes` struct, one entry per tag
$\langle V \mid S \rangle$ (clause)	`Clause` record with `vars` and `stypes`
Clauses (clauses map)	`Clauses` keyed by variable-set
$\mathsf{TType} = (I, C)$	`TType` with `insts` and `clauses`
$\mathsf{refine}^{f,g}$ (filter-map)	`FilterType` transformer
diff (type difference)	`TType::diff`, `Clause::diff`, etc.
$\mathsf{single}(s)$	Single-clause `TType` with one SType
$\mathsf{def}(R, d)$ (defaulting)	`setLocal` with `getSTypes` check

Key implementation behaviors modeled.

1. **Tag-indexed storage**: STypes stores at most one SType per runtime tag.
2. **Same-tag merge**: Union/intersection on STypes merge same-tag alternatives (Definition 11).
3. **Tag-elimination diff**: $\mathsf{diff}_{\mathsf{STypes}}$ removes whole tags, not same-tag subsets.
4. **Top-clause behavior**: Refinement and diff on $\langle \emptyset \mid \top \rangle$ behave specially.
5. **Call-signature modes**: None/Maybe/Definite for callable objects.